

Article

Monitoring of Respiratory Disease Patterns in a Multimicrobially Infected Pig Population Using Artificial Intelligence and Aggregate Samples

Matthias Eddicks ¹, Franziska Feicht ¹, Jochen Beckjunker ², Marika Genzow ², Carmen Alonso ², Sven Reese ³, Mathias Ritzmann ¹ and Julia Stadler ^{1,*}

- ¹ Clinic for Swine at the Centre for Clinical Veterinary Medicine, Ludwig-Maximilians-University München, 85764 München, Germany; m.eddicks@lmu.de (M.E.); franziska.feicht@tieraerzte-wonsees.de (F.F.); m.ritzmann@lmu.de (M.R.)
- ² Boehringer Ingelheim Vetmedica GmbH, Ingelheim, 55216 Ingelheim am Rhein, Germany; jochen.beckjunker@boehringer-ingelheim.com (J.B.); marika.genzow@boehringer-ingelheim.com (M.G.); carmen.alonso@boehringer-ingelheim.com (C.A.)
- ³ Institute for Anatomy, Histology and Embryology, LMU Munich, 80539 Munich, Germany; sven.reese@lmu.de
- * Correspondence: j.stadler@med.vetmed.uni-muenchen.de; Tel.: +49-89-2180-78903

Abstract: A 24/7 AI sound-based coughing monitoring system was applied in combination with oral fluids (OFs) and bioaerosol (AS)-based screening for respiratory pathogens in a conventional pig nursery. The objective was to assess the additional value of the AI to identify disease patterns in association with molecular diagnostics to gain information on the etiology of respiratory distress in a multimicrobially infected pig population. Respiratory distress was measured 24/7 by the AI and compared to human observations. Screening for swine influenza A virus (swIAV), porcine reproductive and respiratory disease virus (PRRSV), *Mycoplasma (M.) hyopneumoniae*, *Actinobacillus (A.) pleuropneumoniae*, and porcine circovirus 2 (PCV2) was conducted using qPCR. Except for *M. hyopneumoniae*, all of the investigated pathogens were detected within the study period. High swIAV-RNA loads in OFs and AS were significantly associated with a decrease in respiratory health, expressed by a respiratory health score calculated by the AI. The odds of detecting PRRSV or *A. pleuropneumoniae* were significantly higher for OFs compared to AS. qPCR examinations of OFs revealed significantly lower Ct-values for swIAV and *A. pleuropneumoniae* compared to AS. In addition to acting as an early warning system, AI gained respiratory health data combined with laboratory diagnostics, can indicate the etiology of respiratory distress.

Keywords: surveillance; sample types; novel; diagnostics; respiratory disease; influenza; oral fluids; bioaerosol samples

1. Introduction

Health monitoring is a crucial element in the production of livestock. Next to animal welfare issues and economical losses [1–5], animal health also depicts a major column for the public health sector. A major challenge for pigs’ health is the porcine respiratory disease complex (PRDC) [6]. This disease complex describes a clinical condition associated with treatment-resistant respiratory disease of growing–finishing pigs of a multifactorial etiology, including infectious and noninfectious factors [7]. Porcine circovirus type 2 (PCV2) [8,9], porcine reproductive and respiratory syndrome virus (PRRSV) [6,10], swine influenza A virus (swIAV) [11,12], and *Mycoplasma (M.) hyopneumoniae* [6,13] are considered as the major pathogens involved in the PRDC [6]. Continuous monitoring of animal behavior or evaluation of clinical signs related to pathological changes in the animals’ health is required to reach and maintain a high health status in swine operations. In recent years, several approaches concerning automated analysis of changes in animal behavior or appearance of



Citation: Eddicks, M.; Feicht, F.; Beckjunker, J.; Genzow, M.; Alonso, C.; Reese, S.; Ritzmann, M.; Stadler, J. Monitoring of Respiratory Disease Patterns in a Multimicrobially Infected Pig Population Using Artificial Intelligence and Aggregate Samples. *Viruses* **2024**, *16*, 1575. <https://doi.org/10.3390/v16101575>

Academic Editor: Aruna Ambagala

Received: 30 August 2024

Revised: 2 October 2024

Accepted: 4 October 2024

Published: 6 October 2024



Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

使用人工智能和空气样本监测多微生物感染猪群的呼吸道疾病模式

马蒂亚斯·埃迪克斯¹, 弗朗齐斯卡·费希特¹, 约亨·贝克琼克², 玛丽卡·根佐夫², 卡门·阿隆索², 斯文·雷瑟³, 马蒂亚斯·里茨曼¹ 和朱莉娅·施塔德勒^{1,*}

- ¹ 慕尼黑路德维希-马克西米利安大学临床兽医学中心猪病诊所, 85764 慕尼黑, 德国; m.eddicks@lmu.de (M.E.); franziska.feicht@tieraerzte-wonsees.de (F.F.); m.ritzmann@lmu.de (M.R.)² 勃林格殷格翰动物保健有限公司, 英格尔海姆, 55216 英格尔海姆河畔, 德国; jochen.beckjunker@boehringer-ingelheim.com (J.B.); marika.genzow@boehringer-ingelheim.com (M.G.); carmen.alonso@boehringer-ingelheim.com (C.A.)³ 慕尼黑路德维希-马克西米利安大学解剖学、组织学和胚胎学研究所, 80539 慕尼黑, 德国; sven.reese@lmu.de* 通讯: j.stadler@med.vetmed.uni-muenchen.de; 电话: +49-89-2180-78903

摘要: 一种基于AI的24/7咳嗽监测系统与口腔液体 (OFs) 和生物气溶胶 (AS) 为基础的呼吸道病原体筛查相结合, 应用于传统猪保育舍。目的是评估AI在结合分子诊断以识别疾病模式方面的附加价值, 从而获取多微生物感染猪群呼吸道困难的病因信息。AI全天候监测呼吸道困难, 并与人工观察结果进行比较。采用qPCR对猪流感A病毒 (swIAV)、猪繁殖与呼吸综合征病毒 (PRRSV)、胸膜肺炎放线杆菌 (*M. hyopneumoniae*)、胸膜肺炎放线杆菌 (*A. pleuropneumoniae*) 和猪圆环病毒2 (PCV2) 进行筛查。除胸膜肺炎放线杆菌 (*M. hyopneumoniae*) 外, 所有研究的病原体在研究期间均被检测到。OFs和AS中高水平的swIAV-RNA与呼吸道健康状况下降显著相关, AI通过计算呼吸道健康评分来表示这种状况。与AS相比, 检测PRRSV或胸膜肺炎放线杆菌 (*A. pleuropneumoniae*) 的几率显著更高。OFs的qPCR检测结果显示, swIAV和胸膜肺炎放线杆菌 (*A. pleuropneumoniae*) 的Ct值与AS相比显著更低。除了作为早期预警系统外, AI结合实验室诊断获得的呼吸道健康数据, 可以指示呼吸道困难的病因。

关键词: 监测; 样本类型; 新型; 诊断; 呼吸道疾病; 流感; 唾液; 生物气溶胶样本



引用: Eddicks, M.; Feicht, F.; Beckjunker, J.; Genzow, M.; Alonso, C.; Reese, S.; Ritzmann, M.; Stadler, J. 使用人工智能和空气样本监测多微生物感染猪群的呼吸道疾病模式. *病毒* **2024**, *16*, 1575. <https://doi.org/10.3390/v16101575>

学术编辑: Aruna Ambagala

收到: 2024年8月30日 修

改: 2024年10月2日 接受:

2024年10月4日 发布:

2024年10月6日



版权: © 2024 由作者所有。许可方 MDPI, 巴塞尔, 瑞士。本文在创意共享署名 (CCBY) 许可条款和条件 (<https://creativecommons.org/licenses/by/4.0/>) 下开放获取。

clinical signs have been published [14–21]. Within this group, wearable devices or sensor technology are assumed to increase animal health and economy in terms of precision livestock farming [20] and thus might even be beneficial concerning human health in terms of the one health strategy of the WHO. In a more concrete way, artificial intelligence (AI) is considered to be helpful to obtain insight into farm infection dynamics [22], detection of clinical signs [14], and decision making [23]. Although automated recognition of animal health-related behavior or clinical signs is thought to be an increasing market [20], the clinical observation of animal health by humans depicts the counterpart and daily work of farmers and veterinarians in the field of livestock farming. Herd inspection often reveals subjectively perceived results unless standardized examinations are carried out. In addition, due to increasing herd sizes, the time for inspection of the animal population or individuals is limited. From this perspective, 24 h automated animal health monitoring seems to be a promising approach that can support the daily work of farmers and covers the time when no human-based inspection is possible. Moreover, it accommodates the possible role of circadian rhythm concerning the inflammatory response on the clinical outcomes of respiratory disease, as reviewed for humans elsewhere [24].

Next to clinical investigations, laboratory diagnostics are needed to gain sufficient information for the implementation of both short-term actions, such as antimicrobial and antiphlogistic treatment, and long-term measures, including vaccinations and management changes. Proper diagnostics and monitoring resemble an important limitation in the control of the porcine respiratory disease complex (PRDC), as the current methods are often labor-intensive, expensive, and based on invasive techniques such as collecting blood samples or sampling the respiratory tract (i.e., nasal swabs, tonsil scratches, tracheobronchial swabs). In the last decade, aggregate samples such as oral fluids (OFs) or environmental samples have gained increasing interest as cost-effective, non-invasive sampling procedures. In particular, OFs are well established for surveillance or monitoring of viral [25,26] or bacterial pathogens [27–29], antimicrobial residues, stress, and biomarkers [30,31]. Next to OFs, bioaerosol samples (AS) have been evaluated for surveillance or monitoring of viral pathogens of humans [32] or viral and bacterial pathogens of animals [33–35]. Concerning porcine pathogens, PCV2, swIAV [36,37], PRRSV, and *M. hyopneumoniae* [38,39] could be detected in bioaerosols under experimental or field conditions. At least for influenza A, virus bioaerosol sampling revealed similar results compared to oral fluids in a farrow-to-feeder facility [37].

The aim of this longitudinal study was to evaluate the additional value of an AI system for monitoring of respiratory distress in a conventional nursery facility combined with qualitative and quantitative detection of PRDC-associated pathogens in OFs and bioaerosol samples.

2. Materials and Methods

2.1. Farm Description

This blinded prospective follow-up study was conducted in a conventional one-site piglet-producing farm in Germany with a known history of recurring respiratory distress. The farm was known to be positive for PRRSV, swIAV, *A. pleuropneumoniae*, and *M. hyopneumoniae* based on PCR analysis for diagnostic investigations performed by the herd attending veterinarian. The piglets were vaccinated against PRRS (UNISTRAIN® PRRS, HIPRA, Amer, Spain, i.m.), porcine circovirus diseases (PCVD) (Ingelvac CircoFLEX®, Boehringer Ingelheim, Ingelheim am Rhein, Germany), *M. hyopneumoniae*-induced diseases (Hyogen®, Ceva, Düsseldorf, Germany) at 21 days of age, and *Lawsonia (L.) intracellularis*-induced diseases (Porcilis® Lawsonia, MSD, Unterschleissheim, Germany) at weaning. Three weeks prior to farrowing, the sows of the corresponding farm were vaccinated with a combined *A. pleuropneumoniae* + *Pasteurella (P.) multocida* and *Streptococcus (S.) suis* autogenous vaccine (SAN VET, Höltinghausen, Germany). In addition, the sows were vaccinated twice a year against swine influenza (Respiporc® FLU3 and Respiporc® FLUpan H1N1,

已发表临床体征 [14–21]。在该群体中，可穿戴设备或传感器技术被认为能够通过精准畜牧业提高动物健康和经济效益 [20]，从而甚至可能有益于人类健康，符合世界卫生组织（WHO）的“同一健康”战略。更具体地说，人工智能（AI）被认为有助于洞察农场感染动态 [22]，临床体征的检测 [14]，以及决策制定 [23]。尽管动物健康相关行为或临床体征的自动化识别被认为是一个不断增长的市场 [20]，但人类对动物健康的临床观察是畜牧业领域农民和兽医的对应工作和日常工作。畜群检查通常揭示主观感知的结果，除非进行标准化检查。此外，由于畜群规模不断扩大，检查动物群体或个体的时间有限。从这个角度来看，24小时自动化动物健康监测似乎是一种有前景的方法，可以支持农民的日常工作，并涵盖无人检查的时间段。此外，它还考虑了昼夜节律对人类呼吸道疾病临床结果中炎症反应的可能作用，正如其他地方对人类进行的综述所述 [24]。

除了临床调查，实验室诊断也需要获取足够的信息，以实施短期行动（如抗菌和抗炎治疗）和长期措施（包括疫苗接种和管理变化）。适当的诊断和监测在控制猪呼吸道疾病综合征（PRDC）方面是一个重要限制，因为当前的方法通常劳动密集、昂贵，并且基于侵入性技术，例如采集血液样本或采样呼吸道（即鼻拭子、扁桃体刮擦、气管支气管拭子）。在过去十年中，聚合样本（如唾液（OFs）或环境样本）作为经济高效的、非侵入性采样程序越来越受到关注。特别是，OFs已广泛用于监测病毒 [25,26] 或细菌病原体 [27–29]，抗菌残留、应激和生物标志物 [30,31]。除了OFs之外，生物气溶胶样本（AS）已被评估用于监测人类病毒病原体 [32]或动物病毒和细菌病原体 [33–35]。关于猪病原体，PCV2、swIAV [36,37]，PRRSV和猪支原体 [38,39] 在实验或现场条件下可能被检测到在生物气溶胶中。至少对于甲型流感，病毒生物气溶胶采样在产房至育肥猪设施 [37]中与唾液监测结果相似。

这项纵向研究的目的是评估人工智能系统在监测传统保育舍中呼吸困难的额外价值，并结合对 OFs 和生物气溶胶样本中 PRDC 相关病原体的定性和定量检测。

2. 材料与方法

2.1. 农场描述

这项盲法前瞻性随访研究是在德国一家有反复呼吸道困难的已知猪场进行的。该农场根据兽医为诊断调查进行的PCR分析，已知呈阳性，包括猪繁殖与呼吸综合征病毒（PRRSV）、猪流感病毒（swIAV）、胸膜肺炎放线杆菌和多杀性巴氏杆菌。猪场在21天龄时对仔猪接种了猪繁殖与呼吸综合征（UNISTRAIN® PRRS，HIPRA，美国，西班牙，肌肉注射）、猪圆环病毒病（PCVD）（Ingelvac CircoFLEX®，勃林格殷格翰，德国英格尔海姆）和多杀性巴氏杆菌诱导的疾病（Hyogen®，Ceva，杜塞尔多夫，德国），并在断奶时接种劳森氏体（*L.*）细胞内菌诱导的疾病（Porcilis® Lawsonia，MSD，德国 Unterschleissheim）。产仔前三周，相应农场的母猪接种了联合胸膜肺炎放线杆菌 + 多杀性巴氏杆菌和猪链球菌自家苗（SAN VET，德国霍尔廷豪森）。此外，母猪每年接种两次猪流感（Respiporc® FLU3和Respiporc® FLUpan H1N1，Ceva，杜塞尔多夫，德国），并每季度接种一次猪繁殖与呼吸综合征（UNISTRAINPRRS，HIPRA，美国，西班牙，皮下注射）。

Ceva, Düsseldorf, Germany) and quarterly against PRRS (UNISTRAIN® PRRS, HIPRA, Amer, Spain, i.d.).

The sow farm was producing piglets in a one-week batch farrowing interval. After weaning at 28 days of age, the pigs were transferred to the corresponding nursery facilities. The nursery unit was managed batch-wise, all-in all-out. The nursery unit was designed to house 600 pigs per airspace and consisted of 20 pens with 25–30 pigs per pen on plastic slatted floors. The animals received a commercial diet via a sensor-controlled liquid feeding trough. Water was available ad libitum via bowl- and nipple-drinkers. The ventilation system consisted of a diffused ceiling for supply air and underfloor extraction for exhaust air. The barns were power washed with foam (MS foam A®, MS Schippers, Kerken, Germany) and disinfected (MS MegaDes Oxy®, MS Schippers, Kerken, Germany) after each batch.

2.2. Study Design and Data Collection

For the present study, two consecutive batches (batch 1 (B1), 520 pigs and batch 2 (B2), 519 pigs) in the nursery were enrolled. The day of placement into the nursery unit was considered as study day 0. The piglets in both batches were monitored for 39 days. All procedures were approved by the internal ethical commission of the Veterinary Faculty of the Ludwig-Maximilians-University München (reference 269-24-04-2021).

2.3. Automated Cough Monitoring

The commercially available AI device (SoundTalks® NV, Leuven, Belgium) performed continuous and automated measurements of coughs and was able to distinguish between a pigs’ cough and other environmental noises in the barn. It compared the previous day’s coughing level with the current one and calculated a numerical score (respiratory health status; ReHS) with values ranging from 0 (worst case) to 100 (best case). The alerts could be recognized through LED lights on the monitor and through the website. A normal ReHS, which ranges from 60 to 100, was represented by a green signal. Yellow and red alerts indicated a potential and a high risk of a respiratory problem and were equivalent to 40–59 and 0–39 ReHS, respectively. Two SoundTalks® boxes (M1, M2) were installed in the alleyway of the barn. Blinding of the investigator was ensured by setting the LEDs of the monitors to “science mode” (blue LED) over the entire study period (Figure 1). The ReHS was obtained daily (24/7) over the entire study period. The total ReHS displayed the lowest score obtained from one of the two monitors.



Figure 1. Left picture: SoundTalks® monitor in science mode (blue led). Right picture: Plugged gateway in science mode (blue led).

Ceva, 杜塞尔多夫, 德国) 和每季度接种一次猪繁殖与呼吸综合征 (UNISTRAIN® PRRS, HIPRA, 美国, 西班牙, 皮下注射)。

该母猪场采用每周一批的分娩间隔生产仔猪。仔猪在28天龄断奶后, 被转移到相应的保育舍设施。保育舍单元采用分批管理、全进全出的方式。该保育舍单元设计为每个空间可容纳600头猪, 由20个栏组成, 每个栏容纳25-30头猪, 采用塑料漏缝地板。动物通过传感器控制的液体饲槽接收商业饲料。自由饮水, 通过碗式和乳头式饮水器提供。通风系统由供风用的扩散式天花板和排风用的地板下抽风系统组成。每批结束后, 畜舍使用泡沫 (MS泡沫A®, MS Schippers, 德国克伦肯) 进行高压清洗, 并使用消毒剂 (MS MegaDes Oxy®, MS Schippers, 德国克伦肯) 进行消毒。

2.2. 研究设计与数据收集

在本研究中, 保育舍中的两个连续批次 (第一批 (B1), 520头仔猪和第二批 (B2), 519头仔猪) 被纳入研究。进入保育舍单元的当天被视为研究第0天。两个批次的仔猪被监测39天。所有程序获得路德维希·马克西米利安大学慕尼黑兽医学院内部伦理委员会的批准 (参考编号269-24-04-2021)。

2.3. 自动咳嗽监测

市售AI设备 (SoundTalks® NV, 比利时鲁汶) 可对咳嗽进行持续和自动测量, 并能区分猪咳嗽与其他猪舍环境噪音。它将前一天的咳嗽水平与当前水平进行比较, 并计算出一个数值评分 (呼吸健康状况; ReHS), 评分范围从0 (最差情况) 到100 (最佳情况)。可通过显示器上的LED灯和网站接收警报。正常的ReHS (范围60至100) 以绿色信号表示。黄色和红色警报分别表示潜在的和高风险的呼吸问题, 相当于40-59和0-39 ReHS。在猪舍的走廊安装了两个SoundTalks® 设备 (M1, M2)。通过在整个研究期间 (图1) 将监控器的LED设置为 “科学模式” (蓝色LED) 来确保研究人员的盲法。在整个研究期间, 每日 (24/7) 获取ReHS。显示的总ReHS

从两个监测器中获得的最低分。



图1. 左图: SoundTalks® 在科学模式下的监控设备 (蓝色LED)。右图: 插入式网关在科学模式下的状态 (蓝色LED)。

2.4. Clinical Cough Monitoring

In addition to the continuously automated AI-based cough monitoring, a clinical coughing score (CCS) was determined as published elsewhere [40]. Briefly, the coughing index was calculated as follows:

$$\text{Coughing index (CI) [\%]} = \frac{\text{Total number of coughing bouts (CC)}}{[\text{number of examined pigs (n)} * \text{total time of observation (min)}] * 100}$$

The score was gathered daily within the first five days after weaning. Subsequently, the clinical monitoring was conducted three times a week (Monday, Wednesday, Friday) in each batch until the end of the observational period on day 39. The coughing score was always determined by the same person (the investigator) at the same time during the morning hours.

2.5. Laboratory Diagnostic Sampling

Four OFs and two AS were collected (Figure 2) on the same days as the clinical coughing score. For the collection of the oral fluids, a cotton rope (BASKO Aleksander Skoracki, Poznań, Poland; length 100 cm, 3 single cords à Ø1 cm, in total ca. Ø3 cm) was placed for approximately 20 min in each pen next to one of the SoundTalks® monitors. The overall sampling procedure and further processing was performed as published elsewhere [41].

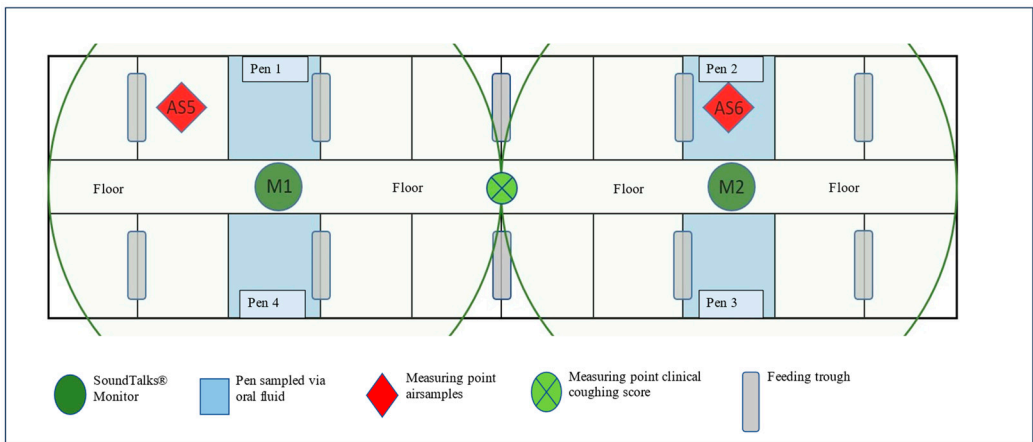


Figure 2. Outline of the study compartment and position of oral fluids and air sample collection points, clinical coughing score assessment, and SoundTalks® monitor (M1 and M2). The radius lines correspond to the coverage range (10 m) of the respective SoundTalks® monitor (M1 and M2).

ASs were obtained using two battery-powered AirPrep™ Cub samplers (AirPrep Model ACD210; InnovaPrep, Drexel, MO, USA) according to the sampling scheme of the clinical coughing score. Prior to sampling, a sterile single-use filter (Ø52 mm) made of a dielectric polymer fiber was inserted into each sampler. In the compartment, both samplers were programmed using an intuitive control panel with a flow rate of 200 L/min and a collection time of 60 min. At the end of the 60 min, and with a filtration of 12 m³, the sampler stopped automatically. One sampler was placed in the front and one in the back of the study compartment (Figure 1, red diamonds (AS5 and AS6)). To ensure that constant measurement points could be used for both Airprep™ cubs, chains were attached to the existing steel beams, and a link was marked. The height of the air sampler was adjusted to the size of the piglets during nursery. In B1, the air inlet opening was located at a height of 77 cm (measured from the floor) from study day 1–26 and at a height of 87 cm from study day 27–39. In B2, the air inlet opening was at a height of 77 cm from study day 1–33 and at a height of 92 cm above the floor from study day 34–39. At the end of the sampling period, the filter was rinsed using the AirPrep Filter and Elution Kit. Briefly, the top of the filter was pressed onto the sample cup, and the elutor adapter was placed on top. Subsequently,

2.4. 临床咳嗽监测

除了持续自动的基于人工智能的咳嗽监测外，还根据文献另行确定了临床咳嗽评分 (CCS) [40]。简而言之，咳嗽指数的计算方法如下：

$$\text{咳嗽指数 (CI) [\%]} = \frac{\text{总咳嗽次数 (CC)}}{[\text{受检猪数量 (n)} * \text{总观察时间 (分钟)}] * 100}$$

断奶后五天内每日收集评分。随后，在每个批次中，临床监测每周进行三次（周一、周三、周五），直至观察期第39天结束。咳嗽评分始终由同一人（研究人员）在早晨同一时间确定。

2.5. 实验室诊断采样

四份OFs和两份空气样本（图2）在同一天收集临床咳嗽评分。用于收集口腔液体，将一根棉绳（BASKO Aleksander Skoracki, 波兹南, 波兰；长度100厘米，3股单绳，直径1厘米，总直径约3厘米）放置在每个猪栏中约20分钟，靠近其中一个SoundTalks®监视器。整体采样程序和进一步处理如其他地方所述 [41]。

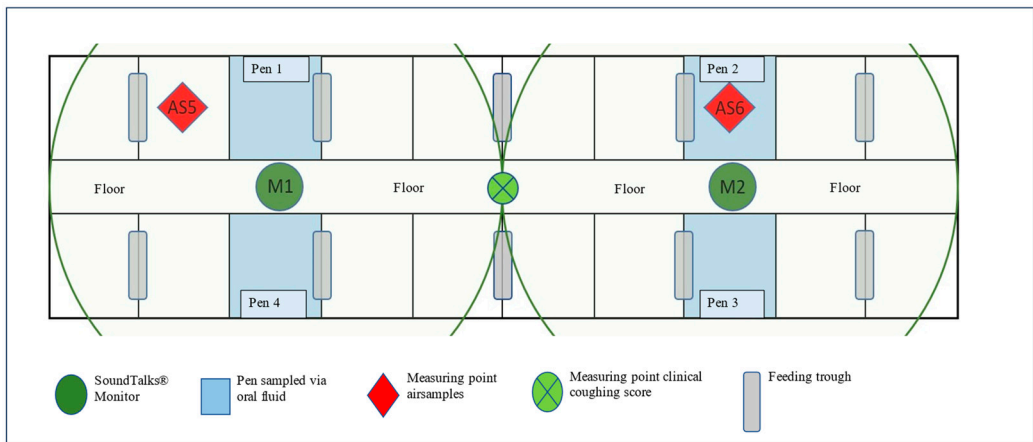


图2. 研究单元的示意图和口腔液体、空气样本采集点、临床咳嗽评分评估以及SoundTalks® 监视器（M1和M2）的位置。半径线对应于相应SoundTalks® 监视器（M1和M2）的覆盖范围（10米）。

空气样本是通过两台电池供电的AirPrep™立方体采样器（AirPrep型号ACD210; InnovaPrep, 德雷塞尔, 密苏里州, 美国）根据临床咳嗽评分的采样方案获得的。采样前，每个采样器中均插入了一个直径52毫米的无菌一次性过滤器（由介电聚合物纤维制成）。在隔间中，两台采样器均使用具有200升/分钟流速和60分钟收集时间的直观控制面板进行编程。在60分钟后，随着12米的过滤，采样器自动停止。一台采样器放置在研究隔间的正面，另一台放置在后面（图1，红色菱形（AS5和AS6））。为了确保两个Airprep立方体可以使用恒定的测量点，链条被连接到现有的钢梁上，并标记了一个连接点。空气采样器的高度在保育舍期间根据仔猪的大小进行调整。在B1中，空气入口位于从第1天到第26天77厘米的高度（从地面测量），从第27天到第39天为87厘米。在B2中，空气入口从第1天到第33天位于77厘米的高度，从第34天到第39天位于地面上92厘米。采样结束时，使用AirPrep过滤器和洗脱试剂盒对过滤器进行冲洗。简而言之，将过滤器的顶部压在样品杯上，并将洗脱适配器放在顶部。随后，³，采样器停止自动。一台采样器放置在研究隔间的正面，另一台放置在后面（图1，红色菱形（AS5和AS6））。为了确保两个Airprep立方体可以使用恒定的测量点，链条被连接到现有的钢梁上，并标记了一个连接点。空气采样器的高度在保育舍期间根据仔猪的大小进行调整。在B1中，空气入口位于从第1天到第26天77厘米的高度（从地面测量），从第27天到第39天为87厘米。在B2中，空气入口从第1天到第33天位于77厘米的高度，从第34天到第39天位于地面上92厘米。采样结束时，使用AirPrep过滤器和洗脱试剂盒对过滤器进行冲洗。简而言之，将过滤器的顶部压在样品杯上，并将洗脱适配器放在顶部。随后，

the filter was washed out with the elution fluid canister containing TRIS medium using a specially developed procedure via wet foam. This foam broke the electrical tension on the filter surface and disintegrated within a few seconds in the sample cup, resulting in 6 mL of wash solution.

The samples were shipped, cooled and promptly processed upon arrival at the laboratory. All assays were performed in a commercial laboratory using accredited assay methods (Table 1). To extract the genomes, we used KingFisher™ Duo Prime (Thermo Fisher Scientific; Waltham, MA, USA) with an ID Gene™ Mag Universal Extraction Kit (Innovative Diagnostics, Grabels, France). Subsequently, the eluates were frozen and stored in KingFisher™ Elution Strips (Thermo Fisher Scientific; Vantaa, Finland) at −20 °C. After the completion of one run, the selected OFs and AS were analyzed collectively. The “BALF” profile provided by the laboratory was used for this purpose, which included the PCR assays listed in Table 1. The real-time PCR was rated as positive if the Ct-value was ≤40.

Table 1. Name and manufacturer of the assays used in the present study.

Pathogen	Name of the Used Assay	Manufacturer
PRRSV	Virotype PRRSV NA/EU	INDICAL BIOSCIENCE GmbH; Leipzig, Germany
PCV2	Virotype PCV2/PCV 3	INDICAL BIOSCIENCE GmbH; Leipzig, Germany
swIAV	Virotype Influenza A RT-PCR	INDICAL BIOSCIENCE GmbH; Leipzig, Germany
<i>M. hyopneumoniae</i>	EXOone <i>Mycoplasma hyopneumoniae</i>	exopol; San Mateo de Gállego, Zaragoza, Spain
<i>A. pleuropneumoniae</i>	EXOone <i>Actinobacillus pleuropneumoniae</i>	exopol; San Mateo de Gállego, Zaragoza, Spain

2.6. Statistical Analysis

The statistical program IBM SPSS Statistics® (version 29.0, IBM® SPSS Inc., Chicago, IL, USA) and Microsoft EXCEL® (version 2019, Microsoft Office, Tulsa, OK, USA) were used for the statistical analysis of the data and to create figures and tables. The significance level was $p < 0.05$. The confidence interval was 0.95.

Continuous data were tested for normal distribution using the Kolmogorov–Smirnov test. Due to non-normal distribution of the data, correlations between CCS, ReHS, the Ct-values of OFs, and the Ct-values of AS were determined using Spearman’s rho. Associations between the non-parametric continuous data Ct-value in dependency of the sample material (OFs and AS) were calculated using the Mann–Whitney U test. Frequency data (detection of pathogens by OFs/AS) were analyzed using cross tables. Fisher’s exact test was used to evaluate the association between the frequency of detection of any pathogen and the sample material. Odds ratios were additionally calculated when appropriate. The level of agreement between OFs and AS for each sampling time was calculated using Cohen’s κ coefficient (κ). Agreement was considered poor if $\kappa \leq 0.2$, fair if $0.21 \leq \kappa \leq 0.4$, moderate if $0.41 \leq \kappa \leq 0.6$, substantial if $0.61 \leq \kappa \leq 0.8$, and good if $\kappa > 0.8$ [42].

3. Results

3.1. Cough Monitoring

The automated AI-based cough monitoring system worked without error, and ReHS could be calculated for every study day. In B1, the first ReHS alert (yellow) was observed on study day 17 only by SoundTalks® monitor 1, located near the entrance, whereas both devices recorded a second alert (yellow) on study day 27 (Figure 3A). In B2, both monitors documented an alert on study day 19; however, monitor 2 already recorded a red alert, whereas monitor 1 showed a yellow alert on day 19 and changed to red on day 20 (Figure 3B).

过滤器使用含有TRIS缓冲液的洗脱液罐和专门开发的程序通过湿泡沫进行清洗。这种泡沫打破了过滤器表面的电张力，在样品杯中几秒钟内分解，导致产生6毫升的洗涤液。

样品在运抵实验室后立即进行冷却和及时处理。所有检测均在一家商业实验室中使用经认证的检测方法（表1）。为了提取基因组，我们使用了KingFisher™ Duo Prime（Thermo Fisher Scientific；马萨诸塞州沃瑟姆，美国）和ID Gene™ Mag通用提取试剂盒（Innovative Diagnostics，法国格布拉尔斯）。随后，洗脱液被冷冻并在KingFisher™洗脱条（Thermo Fisher Scientific；芬兰万塔）中保存于 −20 °C。完成一次运行后，选定的OFs和空气样本被集体分析。实验室提供的“BALF”特征用于此目的，其中包括表1中列出的PCR检测。如果Ct值为 ≤40，则实时PCR被评定为阳性。

表 1。 本研究中使用的检测的名称和制造商。

病原体	所用检测试剂名称	制造商
猪繁殖与呼吸综合征病毒	Virotype 猪繁殖与呼吸综合征病毒 NA/EU	INDICAL BIOSCIENCE GmbH; Leipzig, 德国
PCV2	Virotype PCV2/PCV 3	INDICAL BIOSCIENCE GmbH; 德国莱比锡
swIAV	Virotype 流感 A RT-PCR	INDICAL BIOSCIENCE GmbH; 德国莱比锡
<i>M. hyopneumoniae</i>	EXOone胸膜肺炎放线杆菌	exopol; 沙尔马德雷奥, 萨拉戈萨, 西班牙
胸膜肺炎放线杆菌	EXOone胸膜肺炎棒状杆菌	exopol; 沙尔马德雷奥, 萨拉戈萨, 西班牙

2.6. 统计分析

用于数据统计分析并创建图表的统计程序为IBM SPSS Statistics® (版本29.0, IBM® SPSS Inc., 美国芝加哥, IL)和Microsoft EXCEL® (版本2019, Microsoft Office, 美国俄克拉荷马州塔尔萨, OK), 以及0.95的置信区间。显著性水平为 $p < 0.05$ 。

使用Kolmogorov–Smirnov检验对连续数据进行正态分布检验。由于数据非正态分布，使用Spearman’ s rho确定CCS、ReHS、OFs的Ct值以及AS的Ct值之间的相关性。使用Mann–Whitney U检验计算与样本材料（OFs和AS）相关的非参数连续数据Ct值的关联性。使用交叉表分析频率数据（通过OFs/AS检测病原体）。使用Fisher’ s精确检验评估任何病原体检测频率与样本材料之间的关联性。在适当的情况下，还计算了优势比。使用Cohen’ s κ系数κ计算OFs和AS在每个采样时间的一致性水平。如果 $\kappa \leq 0.2$ ，则认为一致性差；如果 $0.21 \leq \kappa \leq 0.4$ ，则认为一致性一般；如果 $0.41 \leq \kappa \leq 0.6$ ，则认为一致性中等；如果 $0.61 \leq \kappa \leq 0.8$ ，则认为一致性较大；如果 $\kappa > 0.8$ [42]，则认为一致性良好。

3. 结果

3.1. 咳嗽监测

自动化AI咳嗽监测系统运行无差错，ReHS可在每一天研究中计算。在B1组，首个ReHS警报（黄色）仅在研究第17天由SoundTalks监测器1记录，该监测器位于入口附近，而两台设备均在研究第27天记录了第二次警报（黄色）（图3A）。在B2组，两台监测器均在研究第19天记录了警报；然而，监测器2已记录红色警报，而监测器1在第19天显示黄色警报，并在第20天变为红色（图3B）。® 位于入口附近的监测器1，而两台设备均在研究第27天记录了第二次警报（黄色）（图3A）。在B2组，两台监测器均在研究第19天记录了警报；然而，监测器2已记录红色警报，而监测器1在第19天显示黄色警报，并在第20天变为红色（图3B）。

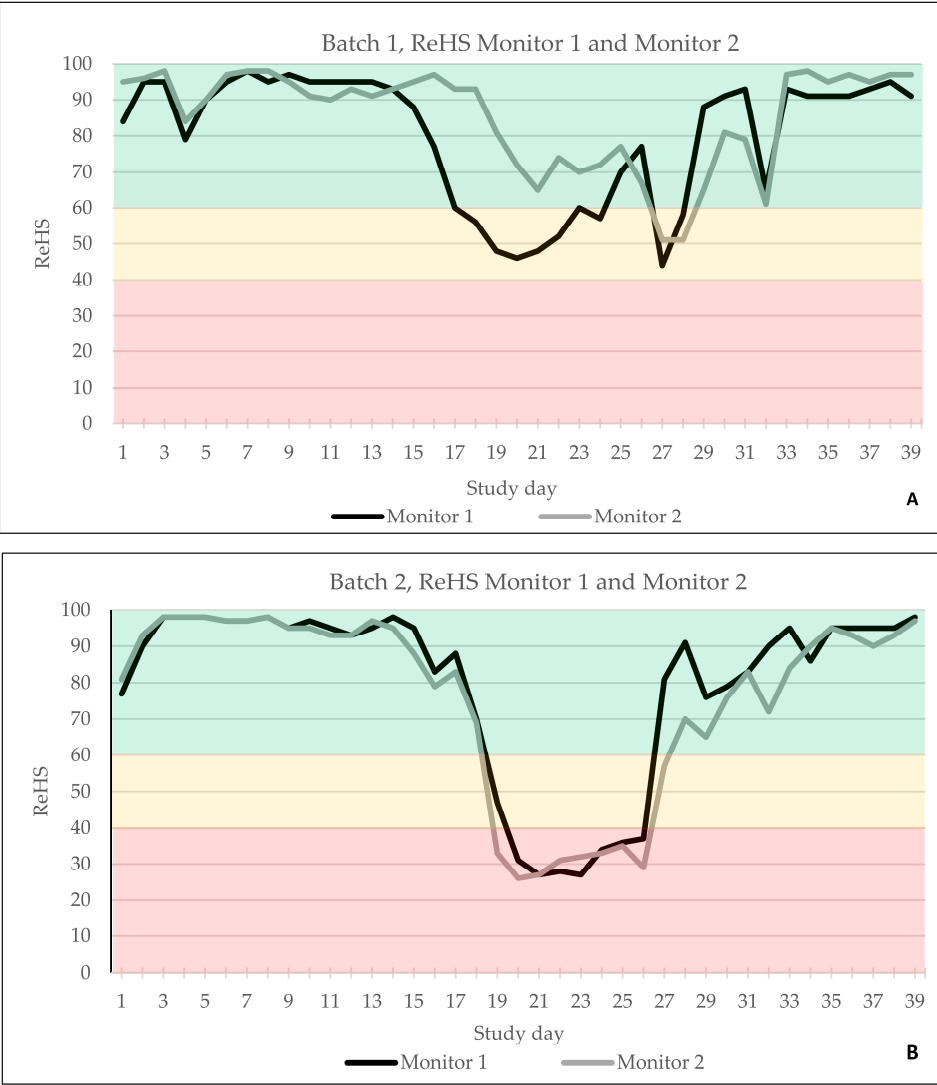


Figure 3. ReHS of batch 1 (A) and batch 2 (B) obtained by monitor 1 and monitor 2 over the entire study period. Green indicated a normal ReHS, which ranges from 60 to 100. Yellow indicated a potential risk of a respiratory problem, equivalent to 40–59 ReHS and red a high risk of a respiratory problem, equivalent to 0–39 ReHS.

The total ReHS from both SoundTalks® monitors and the CCS are displayed in Figure 4A for B1 and in Figure 4B for B2. Briefly, in B1, 29/39 days were classified with a good ReHS (green) and 10/39 days with a moderate ReHS (yellow). In B2, 30/39 days were assigned to a good ReHS (green), 2/39 days to a moderate ReHS (yellow), and 7/39 days to a poor ReHS (red).

Twenty human-conducted clinical coughing scores were obtained in each batch. In B1, only mild clinical signs were recorded by the observer, which did not exceed a coughing index of 0.4%. In B2, more severe coughing was obvious, with a maximum coughing index of 1.26% on study day 20.

Spearman’s rho revealed a significant moderate to medium negative correlation between the ReHS and the CCS for B1 (Spearman’s rho -0.478 ; $p \leq 0.001$) and B2 (Spearman’s rho: -0.468 ; $p \leq 0.001$), indicating that an increase in the CCS was associated with a decrease in the ReHS.

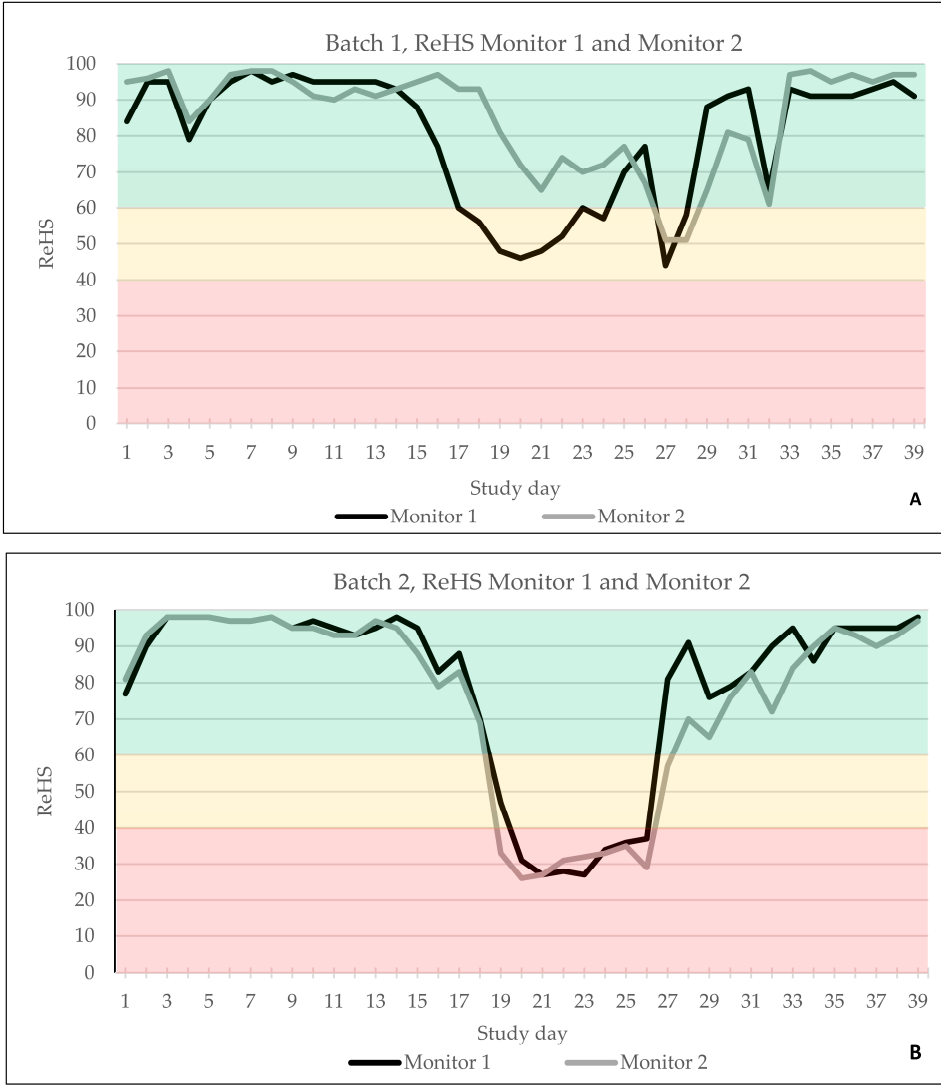


图3. 通过监测1和监测2在整个研究期间获得的批次1 (A) 和批次2 (B) 的呼吸健康评分 (ReHS)。绿色表示正常ReHS，范围在60至100之间。黄色表示潜在的呼吸问题风险，相当于40至59的ReHS，红色表示呼吸问题的高风险，相当于0至39的ReHS。

来自SoundTalks® 监测器和CCS的总ReHS显示在图4A中 (B1) 和图4B中 (B2)。简而言之，在B1中，29/39天被归类为良好ReHS (绿色)，10/39天为中等ReHS (黄色)。在B2中，30/39天被分配为良好ReHS (绿色)，2/39天为中等ReHS (黄色)，7/39天为差ReHS (红色)。

每批次均采集了二十份人类进行的临床咳嗽评分。在B1组中，观察者记录到的仅是轻微的临床症状，咳嗽指数未超过0.4%。而在B2组中，更严重的咳嗽表现明显，研究第20天时咳嗽指数最高达到1.26%。

Spearman’s rho 显示 ReHS 与 CCS 在 B1 (Spearman’s rho -0.478 ; $p \leq 0.001$) 和 B2 (Spearman’s rho: -0.468 ; $p \leq 0.001$) 之间存在显著的中等到中等负相关，表明 CCS 的增加与 ReHS 的减少相关。



Figure 4. Total ReHS and CCS of batch 1 (A) and batch 2 (B) for the observational period of 39 days. Green indicated a normal ReHS, which ranges from 60 to 100. Yellow indicated a potential risk of a respiratory problem, equivalent to 40–59 ReHS and red a high risk of a respiratory problem, equivalent to 0–39 ReHS.

3.2. Molecular Biological Examinations

In total, 240 OFs (120 per batch) and 80 AS (40 per batch) were available for PCR analysis. A detailed overview of the detection rate of each pathogen in the two different batches is presented in Table 2.

Briefly, all of the targeted pathogens, with the exception of *M. hyopneumoniae*, could be detected using PCR in both batches. Significant differences concerning the frequency of detection between OFs and AS could be observed for PRRSV and *A. pleuropneumoniae* in both batches (Table 2). In detail, the odds of detecting PRRSV in OFs were 3.72 times higher (95% CI: 1.96–7.08, $p < 0.001$) compared to AS. Moreover, there were 24.89 higher odds (95% CI: 10.17–60.92, $p < 0.001$) of finding *A. pleuropneumoniae* in OFs using PCR compared to AS. In addition, Cohen’s kappa was calculated to evaluate the agreement between OFs and AS to assign the study population as pathogen-positive or negative for each sample day. Only for swIAV a significant agreement ($p < 0.001$) between OFs and AS with a moderate kappa (0.578) could be observed. For all the other pathogens, no significant agreement could be detected. The median Ct-values of the PCR for swIAV-, PRRSV-, PCV2-, and *A. pleuropneumoniae*-positive OFs and AS are shown using boxplots (Figure 5). RNA or



图4. 总ReHS和CCS (批1 (A) 和批2 (B) 在观察期39天内的值。绿色表示正常ReHS, 范围在60至100; 黄色表示潜在的呼吸问题风险, 相当于40至59 ReHS; 红色表示呼吸问题的高风险, 相当于0至39 ReHS。

3.2. 分子生物学检测

总共, 240 个 OFs (每批 120 个) 和 80 个空气样本 (每批 40 个) 可用于 PCR 分析。每种病原体在两个不同批次中的检测率详细概述于表 2。

简而言之, 除多杀性巴氏杆菌外, 所有目标病原体在两个批次中均可用PCR检测。在两个批次中, PRRSV和胸膜肺炎放线杆菌 在OFs和AS之间的检测频率存在显著差异 (表 2)。具体而言, 在OFs中检测PRRSV的几率比AS高3.72倍 (95% CI: 1.96–7.08, $p < 0.001$)。此外, 在OFs中检测胸膜肺炎放线杆菌 的几率比AS高24.89倍 (95% CI: 10.17–60.92, $p < 0.001$)。此外, 还计算了Cohen’s kappa来评估OFs和AS在将研究人群确定为每个样本日的病原体阳性或阴性方面的协议。仅对于swIAV, OFs和AS之间存在显著协议 ($p < 0.001$), kappa值为中等 (0.578)。对于所有其他病原体, 未检测到显著协议。使用箱线图显示了swIAV、PRRSV、PCV2和胸膜肺炎放线杆菌-阳性OFs和AS的中位数Ct值 (图5)。RNA或

DNA loads of *A. pleuropneumoniae* and swIAV were significantly higher in OFs compared to AS (Figure 5), respectively.

Table 2. Percentage (%) and frequency of detection (n) of specific pathogens using PCR in OFs and AS of batch 1 and batch 2 and for the entire study period, as well as *p*-values in the case of significant differences in the frequency of detection.

	B1			B2			Entire Study Period		
	OFs (n = 64)	AS (n = 32)	<i>p</i> -Value	OFs (n = 60)	AS (n = 30)	<i>p</i> -Value	OFs (n = 124)	AS (n = 62)	<i>p</i> -Value
PRRSV	65.0% (52/80)	32.5% (13/40)	<0.001	27.5% (22/80)	5.0% (2/40)	0.003	46.3% (74/160)	18.8% (15/80)	<0.001
PCV2	7.5% (6/80)	15.0% (6/40)	0.211	5.0% (4/80)	2.5% (1/40)	0.664	6.3% (10/160)	8.8% (7/80)	0.594
swIAV	45.0% (36/80)	57.5% (23/40)	0.246	36.3% (29/80)	35.0% (14/40)	1.00	40.6% (65/160)	46.3% (37/80)	0.410
APP ¹	63.7% (51/80)	7.5% (3/40)	<0.001	70.0% (56/80)	7.5% (3/40)	<0.001	66.9% (107/160)	7.5% (6/80)	<0.001
M. hyo ²	0.0% (0/64)	0.0% (0/32)	-	0.0% (0/60)	0.0% (0/30)	-	0.0% (0/124)	0.0% (0/62)	n.d. *

* n.d.: not detected, ¹ *A. pleuropneumoniae*, ² *M. hyopneumoniae*.

Boxplot of the RT-qPCR for all detected pathogens by material over the entire study period

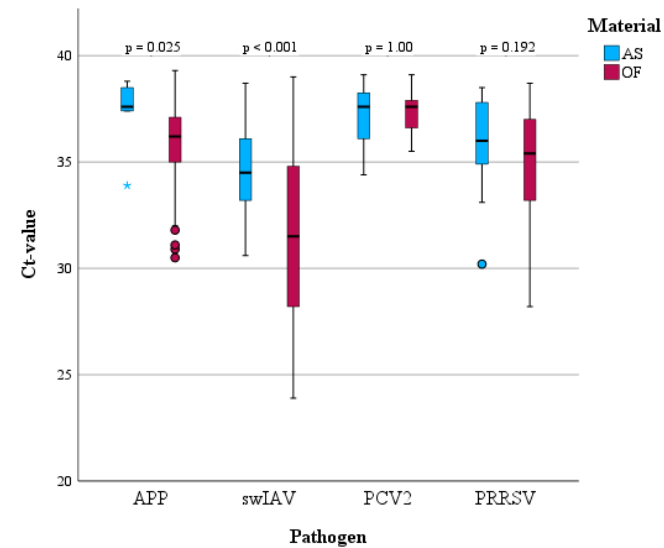


Figure 5. Boxplots (median, lower, and upper quartile, minimum and maximum) of the PCR results for all detected pathogens in OFs and AS over the entire study period.

The Ct-values of positive OFs and AS over the entire study period are shown in Figure 6.

To evaluate potential relationships between the extent of coughing (expressed by ReHS or CCS) and the viral or bacterial loads (expressed by Ct-values) in OFs or AS, bivariate correlations (Spearman’s rho) were calculated. A significant correlation was observed between the Ct-values of the swIAV PCR results and the ReHS (Spearman’s rho: AS: 0.630, *p* < 0.001; OFs: 0.693, *p* < 0.001) and CCS (Spearman’s rho: AS: −0.542, *p* < 0.001, OFs: −0.535, *p* < 0.001) for the entire study period. Figure 7 depict the Ct-values of swIAV for OFs and AS in correlation with the ReHS (Figure 7A) or CCS (Figure 7B), including the regression equation for swIAV Ct-values depending on the used sampling material.

胸膜肺炎放线杆菌和swIAV的DNA负载量在空气样本中显著高于环境样本（图5）， 分别。

表2。使用聚合酶链式反应在批次1和批次2的环境样本和空气样本中检测特定病原体的百分比（%）和频率（n）， 以及整个研究期间以及检测频率显著差异时的p值。

	B1			B2			整个研究期间		
	OFs (n = 64)	AS (n = 32)	<i>p</i> -值	OFs (n = 60)	AS (n = 30)	<i>p</i> -值	OFs (n = 124)	AS (n = 62)	<i>p</i> 值
猪繁殖与呼吸综合征病毒	65.0% (52/80)	32.5% (13/40)	<0.001	27.5% (22/80)	5.0% (2/40)	0.003	46.3% (74/160)	18.8% (15/80)	<0.001
PCV2	7.5% (6/80)	15.0% (6/40)	0.211	5.0% (4/80)	2.5% (1/40)	0.664	6.3% (10/160)	8.8% (7/80)	0.594
swIAV	45.0% (36/80)	57.5% (23/40)	0.246	36.3% (29/80)	35.0% (14/40)	1.00	40.6% (65/160)	46.3% (37/80)	0.410
APP ¹	63.7% (51/80)	7.5% (3/40)	<0.001	70.0% (56/80)	7.5% (3/40)	<0.001	66.9% (107/160)	7.5% (6/80)	<0.001
M. hyo ²	0 0% (0/64) .	0 0% (0/32)	-	0.0% (0/60)	0.0% (0/30)	-	0.0% (0/124)	0.0% (0/62) n.d. *	

* n.d.: 未检出, ¹ 胸膜肺炎放线杆菌, ² 多杀性巴氏杆菌。

Boxplot of the RT-qPCR for all detected pathogens by material over the entire study period

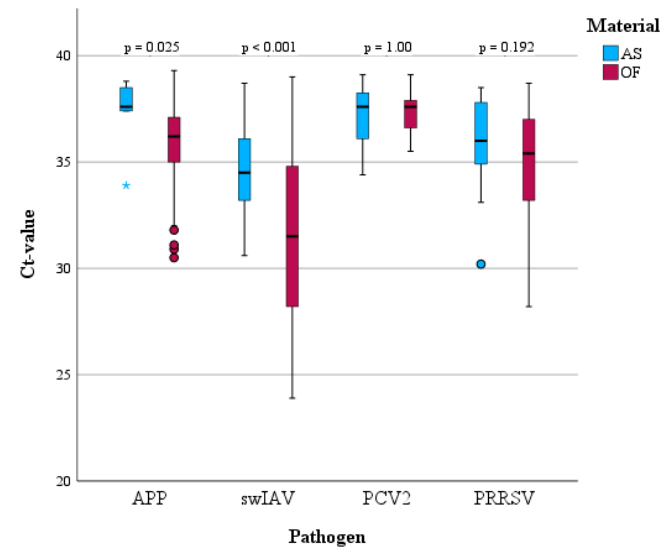


图5。所有检出病原体在空气样本和空气样本中聚合酶链式反应结果的箱线图（中位数、下四分位数和上四分位数、最小值和最大值）在整个研究期间。

整个研究期间阳性空气样本和空气样本的Ct值显示在图6。

为评估咳嗽程度（以ReHS或CCS表示）与空气样本（OFs）或空气样本（AS）中的病毒或细菌负荷（以Ct值表示）之间潜在的关系，计算了双变量相关性（Spearman’ s rho）。在swIAV聚合酶链式反应的Ct值与ReHS (Spearman’ s rho: AS: 0.630, *p* .001; OFs: 0.693, *p* .001) 和CCS (Spearman’ s rho: AS: .542, *p* .001, OFs: .535, *p* .001) 之间存在显著相关性，整个研究期间均如此。图7描绘了与ReHS（图7A）或CCS（图7B）相关的swIAV的OFs和AS的Ct值，包括根据所用采样材料依赖的swIAV Ct值的回归方程。*p* < 0.001; OFs: 0.693, *p* < 0.001)和CCS (Spearman’ s rho: AS: −0.542, *p* < 0.001, OFs:−0.535, *p* < 0.001) 对于整个研究期间。图7 描绘了swIAV的OFs和AS的Ct值与ReHS（图7A）或CCS（图7B）的相关性，包括根据所用采样材料依赖的swIAV Ct值的回归方程。

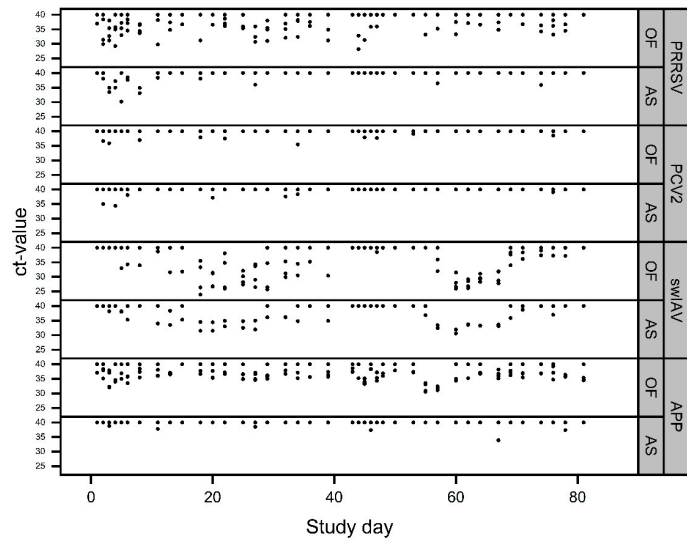


Figure 6. Detection patterns of the different pathogens over the entire study period (batch 1: study days 0–39; batch 2: study days 40–80) in OFs and AS. Ct-values ≥ 40 represent negative results. Overlaying results are possible.

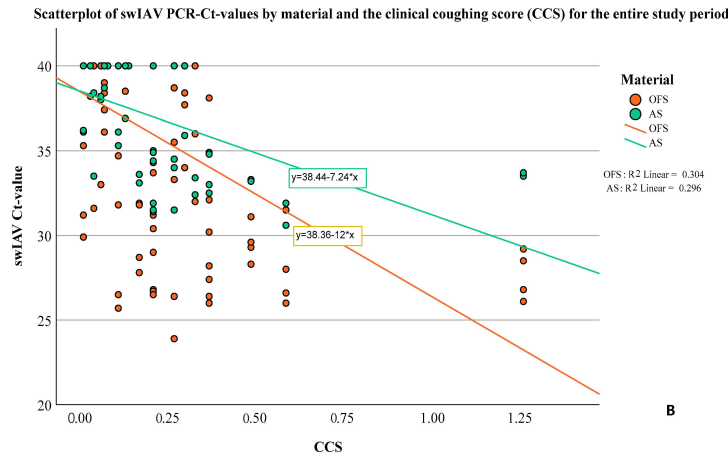
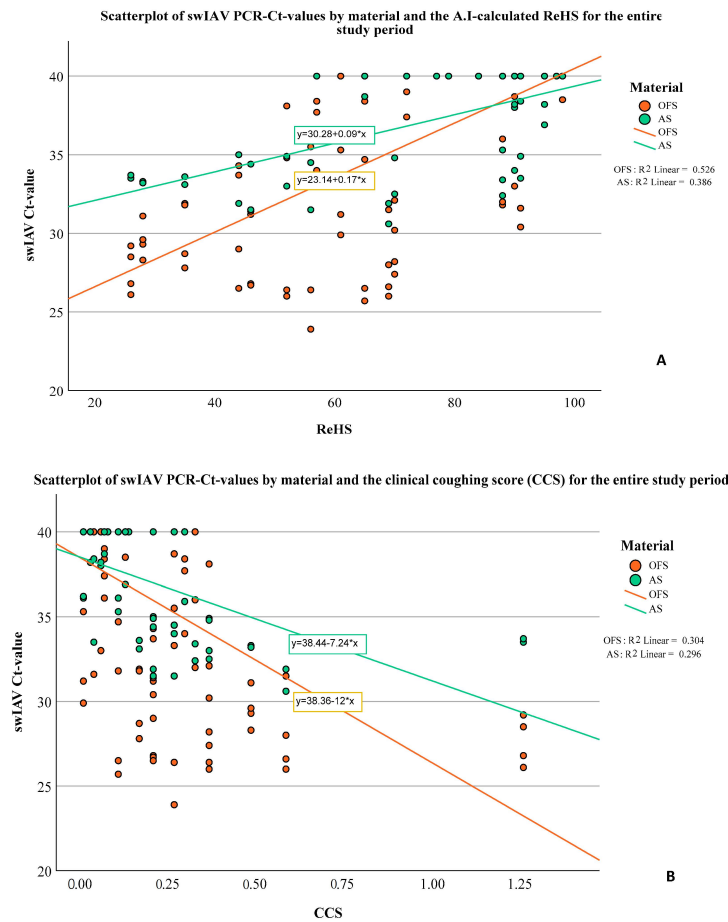


Figure 7. Scatterplot of swIAV PCR Ct-values in correlation with the ReHS (A) or CCS (B) for the entire study period for AS and OFs and the corresponding regression equation.

4. Discussion

Early detection of respiratory distress is essential to preserve animal health and to combat economic losses due to infections with respiratory pathogens. Within the present study, we combined a sound-based AI respiratory health monitoring system with a molecular bio-

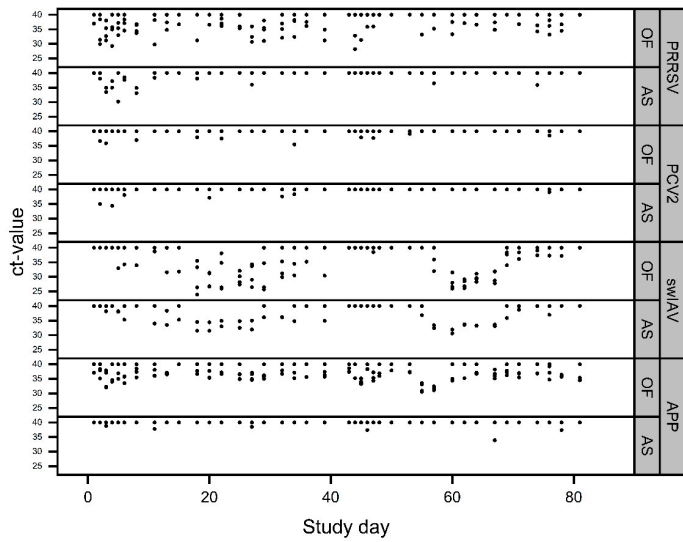


图6. 在整个研究期间（批次1：研究日0-39；批次2：研究日40-80）OFs和AS中不同病原体的检测模式。Ct值 ≥ 40 代表阴性结果。可能存在重叠结果。

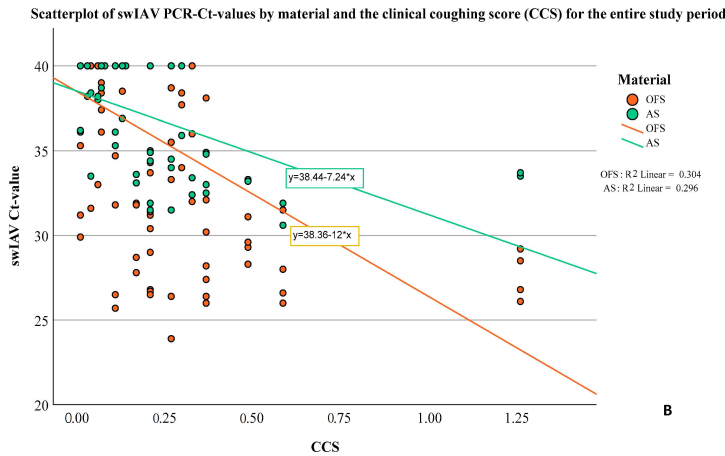
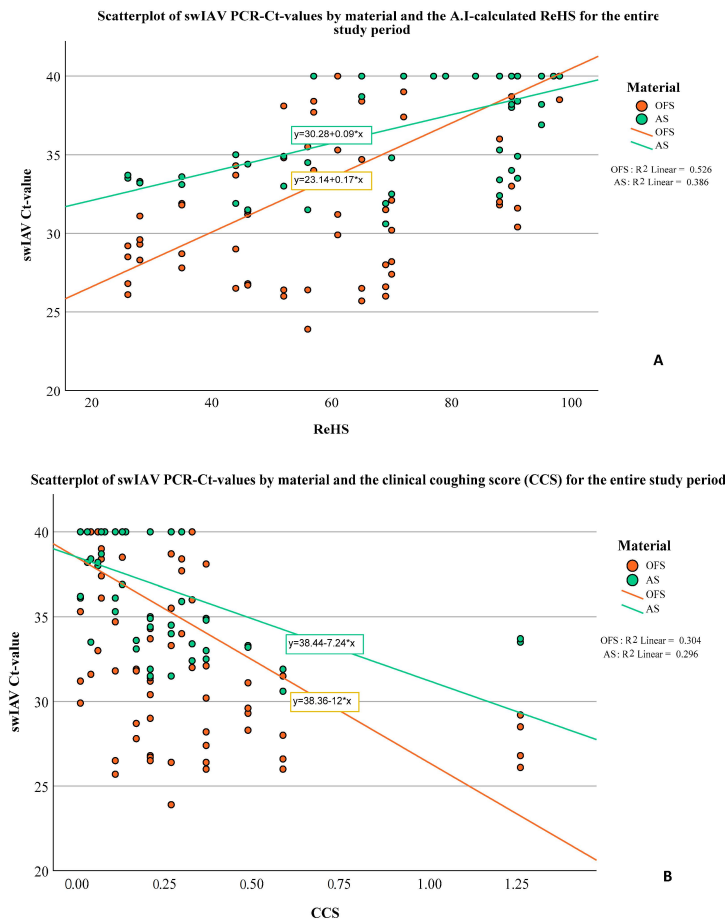


图7. 空气样本中swIAV聚合酶链式反应Ct值与ReHS (A) 或CCS (B) 在整个研究期间的相关散点图以及相应的回归方程。

4. 讨论

早期发现呼吸窘迫对于保护动物健康和应对由呼吸道病原体感染造成的经济损失至关重要。在本研究中，我们将基于声音的人工智能呼吸健康监测系统与分子生物-

logical laboratory screening based on OFs and AS for the detection of pathogens associated with PRDC. Within this scope, we also compared OFs and AS as matrices for the detection of nucleic acids of PRRSV, PCV2, APP, and swIAV. As observed by others [43], the AI-based monitoring system proved to be technically reliable, and respiratory health data could be obtained for each study day. However, limitations concerning the use of this monitoring might include a lack of internet within the range of the stables, which should be checked prior to the installation of such a monitoring system. The comparison between the 24/7 AI and manual coughing monitoring correlated on a moderate to medium level, as also reported by others [16]. In principle, this correlation was expected, as coughing is needed to register a corresponding response by the AI monitoring system. However, the ReHS gave more detailed information, e.g., in the first batch when coughing was present, particularly in the front part of nursery unit. Thus, next to the recognition of clinical signs, AI-based long-term monitoring on farms might also demonstrate localized patterns of disease that indicate issues concerning the farm environment or internal biosecurity. The advantage of continuous AI-based monitoring becomes clearly recognizable as different times of activity of pigs over a daily period can be addressed, whereas a clinical examination is limited in time and only covers the time of the physical presence in the corresponding pig population. This might explain the only moderate correlation between ReHS and CCS in the present examination. The circadian rhythm of the pigs in their surrounding environment influences the times of activity of the pigs [44] and thus the appearance of clinical signs due to the circadian oscillation of the immune system. In terms of respiratory diseases in humans, the severity of clinical signs also shows circadian variability across a 24 h cycle. Increased inflammation and disease severity at night is described for obstructive airway diseases and allergic rhinitis, with the consequence of greater effects of exposure to inflammatory insults at night [24]. Based on a previous investigation, the optimal number of devices for the 600-head barn was determined to be two [45]. However, to ensure optimal sound coverage and to account for variations in infection dynamics, the number of required monitors and the placement of the devices must be thoroughly considered.

The diagnostic screening using PCR revealed the presence of multiple respiratory pathogens in the nursery of the study farm. Thus, the principal preconditions for PRDC were present in the study population. Nevertheless, in order to definitively determine whether a certain combination of pathogens contributed to the clinical signs, pathomorphological examinations would have been required.

Interestingly, a significant correlation was observed between decreasing Ct-values of the swIAV PCR and increasing respiratory distress measured by the AI or the investigator. Thus, an increase in the swIAV viral load in OFs or AS coincided with the extent of the clinical signs in terms of coughing, expressed by reduced ReHS or increased CCS, respectively. Comparable observations were made by Neira et al. [46] who reported a correlation between the quantitative detection of swIAV in oral fluids and the coughing score. Although AI monitoring systems are not able to distinguish between different pathogens, previous initial investigations have shown a bi-modal respiratory distress pattern for swIAV infections in three commercial wean-to-finish facilities [47]. The link between swIAV viral load in OFs or AS and the extent of coughing in our observation also provides evidence to suggest that swIAV might have been the driving force for respiratory distress in this nursery unit. Moreover, based on the comparable pattern of swIAV RNA detection in both batches, an endemic swIAV infection can be assumed. As already postulated by Prost et al. [37] and Anderson et al. [36], no significant differences concerning the qualitative detection of swIAV using PCR were evident between OFs and bioaerosol samples. However, in contrast to Prost et al. [37], swIAV RNA loads were significantly higher in OFs compared to bioaerosol samples.

The detection of PRRSV might either be a result of the MLV vaccination of the piglets or of an infection with a low virulent strain, as no significant correlation between clinical observations and PRRSV detection was observed in the present study. However, further diagnostics to discriminate between field or vaccine strains were not successful due to the

基于OFs和AS的逻辑实验室筛查，用于检测与PRDC相关的病原体。在此范围内，我们还比较了OFs和AS作为检测PRSV、PCV2、APP和swIAV核酸的基质。如其他研究者所观察到的 [43], 基于AI的监测系统被证明在技术上可靠，并且可以为每天的研究日获得呼吸道健康数据。然而，使用此监测系统的局限性可能包括畜舍范围内的互联网缺失，这在安装此类监测系统之前应进行检查。24/7 AI监测与人工咳嗽监测之间的比较达到了中等至一般水平，这也与其他研究者的报告一致 [16]。原则上，这种相关性是预期的，因为咳嗽是注册AI监测系统相应反应所必需的。然而，ReHS提供了更详细的信息，例如在第一批咳嗽存在时，特别是在保育单元的前部。因此，除了识别临床体征外，基于AI的农场长期监测还可能显示疾病的地方性模式，这些模式表明农场环境或内部生物安全存在问题。连续基于AI的监测的优势变得清晰可见，因为可以解决猪在一天中的不同活动时间，而临床检查在时间上有限，并且仅涵盖在相应猪群中的物理存在时间。这可以解释本次检查中ReHS和CCS之间仅中等的相关性。猪在周围环境中的昼夜节律会影响猪的活动时间，从而导致由于免疫系统昼夜节律振荡而出现的临床体征 [44] 在人类呼吸道疾病方面，临床体征的严重程度在24小时周期中也表现出昼夜变化。对于阻塞性气道疾病和过敏性鼻炎，夜间炎症增加和疾病严重程度被描述为，其后果是夜间接触炎症性刺激的影响更大 [24]。根据一项先前的研究，600头猪猪舍的最佳设备数量被确定为两个 [45]。然而，为了确保最佳的声音覆盖范围并考虑感染动态的变化，所需监控器的数量和设备的位置必须仔细考虑。

PCR诊断筛查显示，研究农场保育舍中存在多种呼吸道病原体。因此，研究群体中已具备PRDC的主要先决条件。然而，为了最终确定特定病原体组合是否导致了临床症状，仍需要进行病理形态学检查。

有趣的是，观察到swIAV PCR Ct值下降与AI或研究人员测量的呼吸道 distress增加之间存在显著相关性。因此，swIAV病毒载量在OFs或空气样本中增加，与咳嗽等临床症状的严重程度（分别表现为ReHS降低或CCS增加）相一致。Neira等人 [46] 也报告了口腔液中swIAV定量检测与咳嗽评分之间的相关性。尽管AI监测系统无法区分不同病原体，但之前的初步研究表明，在三个商业保育至出栏设施中，swIAV感染呈现双峰型呼吸道 distress模式 [47]。我们观察到的OFs或空气样本中swIAV病毒载量与咳嗽程度之间的联系，也为swIAV可能是该保育单元呼吸道 distress驱动力的假设提供了证据。此外，基于两批中swIAV RNA检测模式的相似性，可假设存在地方性swIAV感染。正如Prost等人 [37]和Anderson等人 [36], 早已提出的，使用PCR进行swIAV定性检测时，OFs与生物气溶胶样本之间未发现显著差异。然而，与Prost等人 [37], 不同，OFs中的swIAV RNA载量显著高于生物气溶胶样本。

PRRSV 的检测可能是由仔猪接种的 MLV 疫苗或感染低毒力毒株所致，因为在本次研究中未观察到临床观察结果与 PRRSV 检测之间的显著相关性。然而，由于无法区分野毒株或疫苗株的进一步诊断未获成功，

low viral loads in both sampling materials. Although airborne transmission of PRRSV over more than four kilometers was reported [38], the detection rate of viral nucleic acids in bioaerosol samples compared to OFs in our study was rather low. This observation indicates less suitability of bioaerosols for monitoring or surveillance purposes of PRRSV compared to OFs whose sensitivity and suitability to monitor PRRSV were already shown elsewhere [48–50]. Still, these results should be interpreted with caution, because airborne transmission of PRRSV might be strain-dependent, as reported by others [39], and the attenuation of the vaccine strain that was putatively detected here might bias our findings.

Significant advantages of OFs over AS could also be observed for the detection of *A. pleuropneumoniae*. The high sensitivity of OFs concerning the detection of *A. pleuropneumoniae* using PCR was already shown in a previous study under field conditions [27]. The sporadic detection of *A. pleuropneumoniae* DNA in the bioaerosols is in line with the limited capability of *A. pleuropneumoniae* aerosol transmission reported elsewhere [51] and the higher within-pen transmission rate compared to the transmission between different pens [52]. However, the sporadic detection of APP in the bioaerosols indicates the possibility of airborne infections with *A. pleuropneumoniae* within a pig population, as shown by others [51,52]. Moreover, differences in the ventilation rate, air humidity, particle size, and the overall bacterial load in the pig population might influence the detection of bacteria in bioaerosol samples. Particularly, the underfloor extraction for the exhaust air of the stable in the present study might contribute to a reduced amount of bacteria in the air, which could also apply to a certain extent to *A. pleuropneumoniae*. No association between clinical disease patterns and qualitative or quantitative detection of *A. pleuropneumoniae* were obvious in the present study. Possibly, a low virulent strain was circulating in the herd, or the pigs were subclinically infected. However, the applied PCR did not allow for conclusions on the virulence of the circulating strain/s. Moreover, maternally derived immunity, which can last up to 12 weeks of age, might also have prevented clinical disease in our study herd [53].

Concerning PCV2, Harms et al. [8] demonstrated the relevance of the combination of PCV2 and swIAV in a PRDC case series. However, in the present study, the qualitative and quantitative detection rate of PCV2 was low in both sample types. Although Nielsen et al. [54] postulated that PCV2 detection in OFs are not necessarily correlated with clinical signs, PCV2 does not seem to play a major role concerning the clinical outcome in the present nursery, as evident from Figure 5. Moreover, the pigs in the present examination were vaccinated against PCV2. The positive effects of PCV2 vaccination concerning PRDC or PCV2 systemic disease-affected herds [55,56] on clinical outcomes and co-infections are widely known.

Although the farm was categorized as *M. hyopneumoniae* positive and samples were obtained on numerous occasions, *M. hyopneumoniae* DNA was not detected in any of the collected samples. Based on the literature, the detection of *M. hyopneumoniae* DNA from OFs does not appear to have the same sensitivity as the detection of the pathogens mentioned above. This was shown both in studies under experimental conditions and under field conditions. In their challenge test, Pieters et al. [57] examined a wide variety of materials for the detection of *M. hyopneumoniae* DNA. Laryngeal swabs performed best, while OFs were described as occasionally positive, which was already indicated in a previous study [58]. Under field conditions, Hernandez-Garcia et al. [25] recommended several examination time points to compensate for the disadvantages of inconsistent detection. Studies under field conditions at least indicated that OFs at the end of fattening, in association with clinical signs (coughing), appear suitable for the detection of *M. hyopneumoniae* DNA [29]. Furthermore, in the aforementioned study, there was a significant positive correlation between the amount of DNA in the OFs and the extent of lung lesions that could be attributed to enzootic pneumonia. Concerning AS, studies have shown the possibility of detecting *M. hyopneumoniae* DNA in AS under field conditions [38,59]. However, as already mentioned concerning APP, ventilation rate, air humidity, particle size, and bacterial load might influence the detection of bacteria in bioaerosol samples. Particularly, the underfloor

两种采样材料中均存在低病毒载量。[38], 在我们的研究中, 与 OFs 相比, 生物气溶胶样本中病毒核酸的检出率相对较低。这一观察结果表明, 与 OFs 相比, 生物气溶胶在监测或监测 PRRSV 方面的适用性较低, 而 OFs 对监测 PRRSV 的灵敏度和适用性已在其他地方得到证实 [48–50]。尽管如此, 这些结果应谨慎解读, 因为据报道, PRRSV 的气传传播可能具有株依赖性 [39], , 并且这里推定检测到的疫苗株的减弱可能使我们的结果产生偏差。

在检测胸膜肺炎放线杆菌方面, OFs相比AS也具有显著优势。先前一项在田间条件下进行的研究 [27]已经表明, OFs在通过PCR检测胸膜肺炎放线杆菌方面具有高灵敏度。生物气溶胶中偶尔检测到的胸膜肺炎放线杆菌 DNA与据报道的胸膜肺炎放线杆菌 气溶胶传播能力有限 [51]以及与不同猪圈之间传播相比猪圈内部传播率更高 [52]相符。然而, 生物气溶胶中偶尔检测到的APP表明猪群中存在通过空气传播感染胸膜肺炎放线杆菌 的可能性, 正如其他人所展示的那样 [51,52]。此外, 通风率、空气湿度、颗粒大小以及猪群中总体细菌负荷的差异可能会影响生物气溶胶样本中细菌的检测。特别是本研究中畜舍排风的地下抽风可能减少了空气中的细菌量, 这也可能在某种程度上适用于胸膜肺炎放线杆菌。在本研究中, 临床疾病模式与胸膜肺炎放线杆菌的定性或定量检测之间没有明显的关联。可能是群内循环着低毒力菌株, 或者猪只处于亚临床感染状态。然而, 所使用的PCR不允许对循环菌株/的毒力得出结论。此外, 可长达12周持续的母源免疫力也可能阻止了我们研究猪群中的临床疾病 [53]。

关于PCV2, Harms等人 [8] 证明了PCV2和swIAV的组合在PRDC病例系列中的相关性。然而, 在本研究中, 两种样本类型的PCV2定性及定量检出率均较低。尽管 Nielsen等人 [54] 推测PCV2在空气样本中的检出并不一定与临床症状相关, 但PCV2在本保育舍中似乎并未在临床结果中起主要作用, 如图 5所示。此外, 本次检查中的猪只接种了PCV2疫苗。PCV2疫苗接种对PRDC或PCV2全身性疾病影响的群对临床结果和共感染的正效应已广为人知。

尽管农场被归类为<样式 id='1'>猪支原体</样式>阳性, 并且多次采集了样本, 但在所有采集的样本中均未检测到<样式 id='3'>猪支原体</样式>DNA。根据文献, 从空气样本中检测<样式 id='5'>猪支原体</样式>DNA的敏感性似乎与上述病原体的检测敏感性不同。这一点在实验条件和田间条件下进行的研究中均已得到证实。在他们的挑战试验中, Pieters等人 [57]检验了多种材料用于检测<样式 id='8'>猪支原体</样式>DNA。喉拭子效果最佳, 而空气样本被描述为偶尔阳性, 这与先前一项研究 [58]的结论一致。在田间条件下, Hernandez-Garcia等人 [25] 建议设置多个检查时间点, 以弥补检测不一致的缺点。田间条件下的研究表明, 在育肥期末, 结合临床症状 (咳嗽), 空气样本适合检测<样式 id='12'>猪支原体</样式>DNA [29]。此外, 在上述研究中, 空气样本中的DNA含量与可归因于地方流行性肺炎的肺病变程度之间存在显著正相关。关于空气样本, 研究表明在田间条件下有可能检测到<样式 id='15'>猪支原体</样式>DNA [38,59]。然而, 正如之前关于APP所述, 通风率、空气湿度、颗粒大小和细菌负荷可能会影响生物气溶胶样本中细菌的检测。特别是, 本研究所述的排风空气提取过程可能导致了空气中细菌数量的减少。可能的原因是保育舍感染压力较低, 以及研究条件下所用材料的敏感性中等, 导致检测结果为阴性。此外, 下游生产中<样式 id='1'>猪支原体</样式>感染的晚发性和疾病也可能与我们的研究结果相关; 然而, 由于未从育肥猪中采集样本, 我们无法得出结论。

extraction for the exhaust air in the present study might contribute to a reduced amount of bacteria in the air. Likely, a low infection pressure in the nursery, in combination with a medium sensitivity of the used materials under the study conditions, resulted in negative test results. Moreover, a late onset of *M. hyopneumoniae* infection and disease in downstream production might also be of relevance for our findings; however, as no samples were obtained from finishers, we cannot draw a conclusion.

Although aggregate sampling is an accepted tool for monitoring and surveillance of respiratory pathogens, it does not necessarily allow conclusions on the clinical relevance of the detected pathogens. Here, the additional application of an AI respiratory health monitoring system delivers further benefits by indicating the primary etiology of respiratory distress and thus enabling timely diagnostics and treatment. However, further development of the algorithms is needed to allocate coughing patterns to other respiratory pathogens.

5. Conclusions

The AI-based coughing monitoring (SoundTalks®) used in the present examination reliably delivered detailed data on the pigs’ respiratory health and showed localized disease pattern in the nursery. The correlation between the extent of the respiratory distress, expressed by ReHS, and the quantity of swIAV RNA in OFs and AS allowed for speculations on the primary etiology of the respiratory distress. Under the present study conditions, bioaerosol sampling was less sensitive to detecting *A. pleuropneumoniae* and PRRSV using PCR compared to oral fluids. Moreover, *A. pleuropneumoniae* bacterial and swIAV viral loads were significantly higher in oral fluids compared to AS. The application of a long-term AI-based coughing monitoring system in combination with laboratory examinations can be a useful tool to identify relevant pathogens and recurring disease patterns on farms and to recognize respiratory distress under field conditions early.

Author Contributions: Conceptualization, M.E., J.B., M.R. and J.S.; methodology, C.A., M.G. and J.B.; software, F.F., M.E. and J.S.; validation, M.R. and M.E.; formal analysis, F.F., M.E., J.S. and S.R.; investigation, F.F., M.E. and J.S.; resources, M.G., C.A. and J.B.; data curation, F.F., M.E. and J.S.; writing—original draft preparation, F.F., M.E. and J.S.; writing—review and editing, M.R., M.G., J.B., S.R. and C.A.; visualization, F.F., M.E., J.S. and S.R.; supervision, M.R., M.E. and J.S.; project administration, F.F., M.E. and J.S.; funding acquisition, J.B., M.G. and C.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Boehringer Ingelheim Vetmedica GmbH, Ingelheim, Germany (funding number: 646586).

Institutional Review Board Statement: This study was conducted in accordance with the German Animal Welfare Legislation and approved by the internal ethical commission of the Veterinary Faculty of the Ludwig-Maximilians-Universität München (reference 269-24-04-2021).

Informed Consent Statement: Not applicable.

Data Availability Statement: The dataset is available from the corresponding author upon request.

Acknowledgments: We gratefully thank the participating farmer and veterinarian for their valuable support. We would like to thank Dale Polson for providing the air samplers and filtration systems.

Conflicts of Interest: The authors declare no conflicts of interest. J.B., M.G. and C.A. are employees of Boehringer Ingelheim Animal Health, Ingelheim, Germany, which provided funding for this study. The funders were not involved in the collection, analyses, or interpretation of the data.

References

1. McOrist, S. Defining the full costs of endemic porcine proliferative enteropathy. *Vet. J.* **2005**, *170*, 8–9. [[CrossRef](#)] [[PubMed](#)]
2. McOrist, S.; Smith, S.H.; Green, L.E. Estimate of direct financial losses due to porcine proliferative enteropathy. *Vet. Rec.* **1997**, *140*, 579–581. [[CrossRef](#)] [[PubMed](#)]
3. Alarcon, P.; Rushton, J.; Wieland, B. Cost of post-weaning multi-systemic wasting syndrome and porcine circovirus type-2 subclinical infection in England—An economic disease model. *Prev. Vet. Med.* **2013**, *110*, 88–102. [[CrossRef](#)] [[PubMed](#)]

很可能，保育舍中较低的感染压力，加上研究条件下所用材料的敏感性中等，导致了阴性检测结果。此外，下游生产中<样式 id='1'>猪支原体</样式>感染的晚发性和疾病也可能与我们的研究结果相关；然而，由于未从育肥猪中采集样本，我们无法得出结论。

尽管聚合样本是监测和调查呼吸道病原体的公认工具，但它并不一定允许对检测到的病原体作出临床相关性的结论。在此，应用人工智能呼吸道健康监测系统可进一步提供帮助，通过指示呼吸道困难的初始病因，从而实现及时诊断和治疗。然而，需要进一步开发算法，以将咳嗽模式分配给其他呼吸道病原体。

5. 结论

基于人工智能的咳嗽监测（SoundTalks®）在本研究中可靠地提供了关于猪呼吸道健康的详细数据，并在保育舍中显示出局部疾病模式。呼吸道困难程度（由ReHS表示）与OFs和AS中swIAV RNA数量之间的相关性，允许对呼吸道困难的初始病因进行推测。在本研究条件下，与口腔液体相比，生物气溶胶采样使用PCR检测胸膜肺炎放线杆菌和猪繁殖与呼吸综合征病毒灵敏度较低。此外，与AS相比，口腔液体中胸膜肺炎放线杆菌细菌负荷和swIAV病毒负荷显著更高。在农场识别相关病原体 and 反复出现的疾病模式，以及早期识别现场条件下的呼吸道困难的长期基于人工智能的咳嗽监测系统与实验室检查相结合的应用，可作为一个有用工具。

作者贡献：概念化，M.E.、J.B.、M.R.和J.S.；方法学，C.A.、M.G.和J.B.；软件，F.F.、M.E.和J.S.；验证，M.R.和M.E.；形式分析，F.F.、M.E.、J.S.和S.R.；研究，F.F.、M.E.和J.S.；资源，M.G.、C.A.和J.B.；数据管理，F.F.、M.E.和J.S.；初稿撰写，F.F.、M.E.和J.S.；审阅与编辑，M.R.、M.G.、J.B.、S.R.和C.A.；可视化，F.F.、M.E.、J.S.和S.R.；指导，M.R.、M.E.和J.S.；项目管理，F.F.、M.E.和J.S.；资金获取，J.B.、M.G.和C.A.所有作者均已阅读并同意手稿的发表版本。

资助： 本研究由勃林格殷格翰动物保健有限公司（德国英戈尔海姆，资助编号：646586）提供资助。

机构审查委员会声明： 本研究符合德国动物福利法规，并经路德维希-马克西米利安大学慕尼黑兽医学院内部伦理委员会批准（参考编号 269-24-04-2021）。

知情同意声明： 不适用。

数据可用性声明： 数据集可向通讯作者申请获取。 t.

致谢： 我们衷心感谢参与研究的农民和兽医提供的宝贵支持。我们感谢戴尔·波尔森提供了空气样本采集器和过滤系统。

利益冲突声明： 作者声明无利益冲突。J.B.、M.G.和C.A.是勃林格殷格翰动物保健有限公司（德国英戈尔海姆）的员工，该公司为本研究提供资助。资助方未参与数据的收集、分析或解释。

参考文献

1. McOrist, S. Defining the full costs of endemic porcine proliferative enteropathy. *Vet. J.* **2005**, *170*, 8–9. [[CrossRef](#)] [[PubMed](#)]
2. McOrist, S.; Smith, S.H.; Green, L.E. Estimate of direct financial losses due to porcine proliferative enteropathy. *Vet. Rec.* **1997**, *140*, 579–581. [[CrossRef](#)] [[PubMed](#)]
3. Alarcon, P.; Rushton, J.; Wieland, B. Cost of post-weaning multi-systemic wasting syndrome and porcine circovirus type-2 subclinical infection in England—An economic disease model. *Prev. Vet. Med.* **2013**, *110*, 88–102. [[CrossRef](#)] [[PubMed](#)]

4. Renken, C.; Nathues, C.; Swam, H.; Fiebig, K.; Weiss, C.; Eddicks, M.; Ritzmann, M.; Nathues, H. Application of an economic calculator to determine the cost of porcine reproductive and respiratory syndrome at farm-level in 21 pig herds in Germany. *Porcine Health Manag.* **2021**, *7*, 3. [\[CrossRef\]](#)

5. Boeters, M.; Garcia-Morante, B.; van Schaik, G.; Segales, J.; Rushton, J.; Steeneveld, W. The economic impact of endemic respiratory disease in pigs and related interventions—A systematic review. *Porcine Health Manag.* **2023**, *9*, 45. [\[CrossRef\]](#) [\[PubMed\]](#)

6. Thacker, E.L. Immunology of the porcine respiratory disease complex. *Vet. Clin. N. Am. Food Anim. Pract.* **2001**, *17*, 551–565. [\[CrossRef\]](#)

7. Eddicks, M.; Eddicks, L.; Stadler, J.; Hermanns, W.; Ritzmann, M. Der Porcine Respiratory Disease Complex (PRDC)—eine klinische Übersicht. *Tierärztl. Praxis Ausgabe G Großtiere/Nutztiere* **2021**, *49*, 120–132. [\[CrossRef\]](#)

8. Harms, P.A.; Halbur, P.G.; Sorden, S.D. Three cases of porcine respiratory disease complex associated with porcine circovirus type 2 infection. *J. Swine Health Prod.* **2002**, *10*, 27–30. [\[CrossRef\]](#) [\[PubMed\]](#)

9. Kim, J.; Chung, H.K.; Chae, C. Association of porcine circovirus 2 with porcine respiratory disease complex. *Vet. J.* **2003**, *166*, 251–256. [\[CrossRef\]](#) [\[PubMed\]](#)

10. Opriessnig, T.; Gimenez-Lirola, L.G.; Halbur, P.G. Polymicrobial respiratory disease in pigs. *Anim. Health Res. Rev.* **2011**, *12*, 133–148. [\[CrossRef\]](#) [\[PubMed\]](#)

11. Deblanc, C.; Simon, G. Involvement of swine influenza A viruses in the porcine respiratory disease complex. *Virologie* **2017**, *21*, 225–238.

12. Schmidt, C.; Cibulski, S.P.; Andrade, C.P.; Teixeira, T.F.; Varela, A.P.; Scheffer, C.M.; Franco, A.C.; de Almeida, L.L.; Roehe, P.M. Swine influenza virus and association with the porcine respiratory disease complex in pig farms in southern brazil. *Zoonoses Public Health* **2016**, *63*, 234–240. [\[CrossRef\]](#) [\[PubMed\]](#)

13. Thacker, E.L.; Halbur, P.G.; Ross, R.F.; Thanawongnuwech, R.; Thacker, B.J. *Mycoplasma hyopneumoniae* potentiation of porcine reproductive and respiratory syndrome virus-induced pneumonia. *J. Clin. Microbiol.* **1999**, *37*, 620–627. [\[CrossRef\]](#) [\[PubMed\]](#)

14. Chung, Y.; Oh, S.; Lee, J.; Park, D.; Chang, H.H.; Kim, S. Automatic detection and recognition of pig wasting diseases using sound data in audio surveillance systems. *Sensors* **2013**, *13*, 12929–12942. [\[CrossRef\]](#)

15. Pessoa, J.; Rodrigues da Costa, M.; García Manzanilla, E.; Norton, T.; McAloon, C.; Boyle, L. Managing respiratory disease in finisher pigs: Combining quantitative assessments of clinical signs and the prevalence of lung lesions at slaughter. *Prev. Vet. Med.* **2021**, *186*, 105208. [\[CrossRef\]](#) [\[PubMed\]](#)

16. Silva, A.; Storino, G.Y.; Ferreyra, F.S.M.; Zhang, M.; Fano, E.; Polson, D.; Wang, C.; Derscheid, R.J.; Zimmerman, J.J.; Clavijo, M.J.; et al. Cough associated with the detection of *Mycoplasma hyopneumoniae* DNA in clinical and environmental specimens under controlled conditions. *Porcine Health Manag.* **2022**, *8*, 6. [\[CrossRef\]](#)

17. Wang, S.; Jiang, H.; Qiao, Y.; Jiang, S.; Lin, H.; Sun, Q. The Research Progress of Vision-Based Artificial Intelligence in Smart Pig Farming. *Sensors* **2022**, *22*, 6541. [\[CrossRef\]](#) [\[PubMed\]](#)

18. Jorquera-Chavez, M.; Fuentes, S.; Dunshea, F.R.; Warner, R.D.; Poblete, T.; Morrison, R.S.; Jongman, E.C. Remotely Sensed Imagery for Early Detection of Respiratory Disease in Pigs: A Pilot Study. *Animals* **2020**, *10*, 451. [\[CrossRef\]](#) [\[PubMed\]](#)

19. Neethirajan, S. Affective State Recognition in Livestock-Artificial Intelligence Approaches. *Animals* **2022**, *12*, 759. [\[CrossRef\]](#) [\[PubMed\]](#)

20. Neethirajan, S. Recent advances in wearable sensors for animal health management. *Sens. Bio-Sens. Res.* **2017**, *12*, 15–29. [\[CrossRef\]](#)

21. Arulmozhi, E.; Bhujel, A.; Moon, B.E.; Kim, H.T. The Application of Cameras in Precision Pig Farming: An Overview for Swine-Keeping Professionals. *Animals* **2021**, *11*, 2343. [\[CrossRef\]](#)

22. Esener, N.; Green, M.J.; Emes, R.D.; Jowett, B.; Davies, P.L.; Bradley, A.J.; Dottorini, T. Discrimination of contagious and environmental strains of *Streptococcus uberis* in dairy herds by means of mass spectrometry and machine-learning. *Sci. Rep.* **2018**, *8*, 17517. [\[CrossRef\]](#) [\[PubMed\]](#)

23. Ezanno, P.; Picault, S.; Beaune, G.; Bailly, X.; Munoz, F.; Duboz, R.; Monod, H.; Guegan, J.F. Research perspectives on animal health in the era of artificial intelligence. *Vet. Res.* **2021**, *52*, 40. [\[CrossRef\]](#) [\[PubMed\]](#)

24. Comas, M.; Gordon, C.J.; Oliver, B.G.; Stow, N.W.; King, G.; Sharma, P.; Ammit, A.J.; Grunstein, R.R.; Phillips, C.L. A circadian based inflammatory response—implications for respiratory disease and treatment. *Sleep. Sci. Pract.* **2017**, *1*, 18. [\[CrossRef\]](#)

25. Hernandez-García, J.; Robben, N.; Magnée, D.; Eley, T.; Dennis, I.; Kayes, S.M.; Thomson, J.R.; Tucker, A.W. The use of oral fluids to monitor key pathogens in porcine respiratory disease complex. *Porcine Health Manag.* **2017**, *3*, 7. [\[CrossRef\]](#) [\[PubMed\]](#)

26. Ramirez, A.; Wang, C.; Prickett, J.R.; Pogranichniy, R.; Yoon, K.J.; Main, R.; Johnson, J.K.; Rademacher, C.; Hoogland, M.; Hoffmann, P.; et al. Efficient surveillance of pig populations using oral fluids. *Prev. Vet. Med.* **2012**, *104*, 292–300. [\[CrossRef\]](#) [\[PubMed\]](#)

27. Kleinmans, M.; Fiebig, K.; Tabeling, R.; Swam, H.; Duivelshof-Crienen, A.; Ritzmann, M.; Eddicks, M. Explorative Field Study on the Use of Oral Fluids for the Surveillance of *Actinobacillus pleuropneumoniae* Infections in Fattening Farms by an Apx-Real-Time PCR. *Vet. Sci.* **2022**, *9*, 552. [\[CrossRef\]](#)

28. Schott, F.; Hoffmann, K.; Sarno, E.; Bangerter, P.D.; Stephan, R.; Overesch, G.; Haessig, M.; Sidler, X.; Graage, R. Evaluation of oral fluids for surveillance of foodborne and zoonotic pathogens in pig farms. *J. Vet. Diagn. Investig.* **2021**, *33*, 655–663. [\[CrossRef\]](#) [\[PubMed\]](#)

4. 伦肯, C; 纳特斯, C; 斯瓦姆, H; 菲比格, K; 韦斯, C; 艾迪克斯, M; 里茨曼, M; 纳特斯, H. 应用经济计算器确定德国21个猪群农场层面的猪繁殖与呼吸综合征成本. 猪健康管理 2021, 7, 3. [交叉引用]5. 博伊特斯, M; 加西亚-莫兰特, B; 范沙伊克, G; 塞加莱斯, J; 鲁斯通, J; 斯滕内弗尔德, W. 猪地方性呼吸道疾病和相关干预措施的经济影响—系统评价. 猪健康管理 **2023**, *9*, 45 [][]. 交叉引用 PubMed 。 萨克, E.L. 猪呼吸道疾病综合征的免疫学. 兽医学临床. 食品动物实践 2001, 17, 551–565.[交叉引用]7. 艾迪克斯, M; 艾迪克斯, L; 施塔德勒, J; 赫尔曼斯, W; 里茨曼, M. 猪呼吸道疾病综合征 (PRDC) —临床综述. 兽医实践. 大型动物/经济动物 2021, 49, 120–132. [交叉引用]8. 哈姆斯, P.A; 哈尔伯, P.G; 索登, S.D. 三例与猪圆环病毒2感染相关的猪呼吸道疾病综合征. 猪健康生产 2002, 10, 27–30. [交叉引用] [\[PubMed\]](#)9. 金, J; 钟, H.K; 蔡, C. 猪圆环病毒2与猪呼吸道疾病综合征的关联. 兽医学杂志 2003, 166, 251–256. [交叉引用] [\[PubMed\]](#)10. 奥普里森西格, T; 吉梅内斯-利罗拉, L.G; 哈尔伯, P.G. 猪的多重微生物呼吸道疾病. 动物健康研究评论 2011, 12, 133–148. [交叉引用] [\[PubMed\]](#)11. 德布兰, C; 西蒙, G. 猪流感A病毒在猪呼吸道疾病综合征中的作用. 病毒学 2017, 21, 225–238. 12. 施密特, C; 西布爾斯基, S.P; 安德拉德, C.P; 特谢拉, T.F; 瓦雷拉, A.P; 谢弗, C.M; 弗朗科, A.C; 德阿尔梅达, L.L; 罗赫, P.M. 猪流感病毒及其与巴西南部猪场猪呼吸道疾病综合征的关联. 动物公共卫生 2016, 63, 234–240. [交叉引用] [\[PubMed\]](#)13. 萨克, E.L; 哈尔伯, P.G; 罗斯, R.F; 坦纳万古韦奇, R; 萨克, B.J. 支原体肺炎增强猪繁殖与呼吸综合征病毒诱导的肺炎. 临床微生物学杂志 1999, 37, 620–627. [交叉引用] [\[PubMed\]](#)14. 钟毅; 吴; 李; 朴; 昌; 金. 使用音频监控系统中声音数据自动检测和识别猪消耗性疾病. 传感器 2013, 13, 12929–12942. [交叉引用]15. 佩索阿, J; 罗德里格斯-达-科斯塔, M; 加西亚-曼萨尼利亚, E; 诺顿, T; 麦克劳恩, C; 博伊尔, L. 管理育肥猪呼吸道疾病: 宰前临床体征定量评估与屠宰时肺病变流行率的结合. 预防兽医学 2021, 186, 105208. [交叉引用] [\[PubMed\]](#)16. 西尔瓦, A; 斯托里诺, G.Y; 费雷拉, F.S.M; 张, M; 法诺, E; 波尔森, D; 王, C; 德尔斯海德, R.J; 齐默曼, J.J; 克拉维霍, M.J; 等. 控制条件下临床和环境样本中支原体肺炎DNA检测与咳嗽相关. 猪健康管理 2022, 8, 6. [交叉引用]17. 王, S; 姜; 乔; 姜; 林; 孙. 基于视觉的智能养猪中人工智能的研究进展. 传感器 2022, 22, 6541. [交叉引用] [\[PubMed\]](#)18. 约奎拉-切维斯, M; 富恩特斯, S; 邓谢亚, F.R; 沃纳, R.D; 波布莱特, T; 莫里森, R.S; 琼曼, E.C. 遥感影像用于猪呼吸道疾病的早期检测: 一项试点研究. 动物 2020, 10, 451. [交叉引用] [\[PubMed\]](#)19. 尼提拉贾南, S. 动物情绪状态识别—人工智能方法. 动物 2022, 12, 759. [交叉引用] [\[PubMed\]](#)20. 尼提拉贾南, S. 动物健康管理可穿戴传感器最新进展. 生物传感器研究 2017, 12, 15–29.[交叉引用]21. 阿鲁莫吉, E; 布胡杰, A; 穆恩, B.E; 金, H.T. 精准养猪中相机的应用: 养猪专业人员的概述. 动物 2021, 11, 2343. [交叉引用]22. 埃森, N; 格林, M.J; 埃姆斯, R.D; 乔伊特, B; 戴维斯, P.L; 布拉德利, A.J; 多托里尼, T. 通过质谱和机器学习区分奶牛场传染性和环境性的链球菌属牛链球菌菌株. 科学报告 2018, 8, 17517. [交叉引用] [\[PubMed\]](#)23. 埃赞诺, P; 皮卡尔, S; 贝内, G; 巴利, X; 穆诺兹, F; 杜波, R; 莫诺, H; 盖甘, J.F. 人工智能时代动物健康研究展望. 兽医学 2021, 52, 40. [交叉引用] [\[PubMed\]](#)24. 科马斯, M; 戈登, C.J; 奥利弗, B.G; 斯托, N.W; 金, G; 夏尔马, P; 阿米特, A.J; 格鲁宁施泰因, R.R; 菲利普斯, C.L. 基于昼夜节律的炎症反应—对呼吸道疾病和治疗的影响. 睡眠科学实践 2017, 1, 18. [交叉引用]25. 埃尔南德斯-加西亚, J; 罗宾, N; 马涅, D; 埃利, T; 丹尼斯, I; 凯斯, S.M; 汤普森, J.R; 塔克, A.W. 使用口腔液体监测猪呼吸道疾病综合征中的关键病原体. 猪健康管理 2017, 3, 7. [交叉引用] [\[PubMed\]](#)26. 雷蒙德, A; 王; 普里克特, J.R; 波格拉尼奇尼, R; 尹; 金; 约翰逊, J.K; 拉登巴赫, C; 霍格兰德, M; 霍夫曼, P; 等. 使用口腔液体有效监测猪群. 预防兽医学 2012, 104, 292–300. [交叉引用] [\[PubMed\]](#)27. 克莱因曼斯, M; 菲比格, K; 塔贝尔宁, R; 斯瓦姆, H; 杜伊夫霍夫-克里尼恩, A; 里茨曼, M; 艾迪克斯, M. 关于使用口腔液体在育肥农场通过apx-实时pcr监测放线杆菌胸膜肺炎感染的探索性现场研究. 兽医科学 2022, 9, 552. [交叉引用]28. 施奥特, F; 霍夫曼, K; 萨诺, E; 班格特, P.D; 斯蒂芬, R; 奥弗施泰因, G; 哈塞格, M; 西德勒, X; 格拉格, R. 评估口腔液体监测猪场食源性和人畜共患病原体. 兽医诊断研究杂志 2021, 33, 655–663. [交叉引用] [\[PubMed\]](#)

29. Deffner, P.; Maurer, R.; Cvjetkovic, V.; Sipos, W.; Krejci, R.; Ritzmann, M.; Eddicks, M. Cross-sectional study on the in-herd prevalence of *Mycoplasma hyopneumoniae* at different stages of pig production. *Vet. Rec.* **2022**, *191*, e1317. [\[CrossRef\]](#)

30. Giergiel, M.; Olejnik, M.; Jabłoński, A.; Posyniak, A. The Markers of Stress in Swine Oral Fluid. *J. Vet. Res.* **2021**, *65*, 487–495. [\[CrossRef\]](#)

31. Gajda, A.; Jablonski, A.; Gbylik-Sikorska, M.; Posyniak, A. Correlation between oral fluid and plasma oxytetracycline concentrations after intramuscular administration in pigs. *J. Vet. Pharmacol. Ther.* **2017**, *40*, e39–e44. [\[CrossRef\]](#) [\[PubMed\]](#)

32. Tedeschini, E.; Pasqualini, S.; Emiliani, C.; Marini, E.; Valecchi, A.; Laoreti, C.; Ministrini, S.; Camilloni, B.; Castronari, R.; Patoia, L.; et al. Monitoring of indoor bioaerosol for the detection of SARS-CoV-2 in different hospital settings. *Front. Public Health* **2023**, *11*, 1169073. [\[CrossRef\]](#)

33. Song, L.; Wang, C.; Jiang, G.; Ma, J.; Li, Y.; Chen, H.; Guo, J. Bioaerosol is an important transmission route of antibiotic resistance genes in pig farms. *Environ. Int.* **2021**, *154*, 106559. [\[CrossRef\]](#)

34. Nguyen, N.T.T.; Luu, Y.T.H.; Hoang, T.D.; Nguyen, H.X.; Dao, T.D.; Bui, V.N.; Gray, G.C. An epidemiological study of *Streptococcus suis* prevalence among swine at industrial swine farms in Northern Vietnam. *One Health* **2021**, *13*, 100254. [\[CrossRef\]](#) [\[PubMed\]](#)

35. Anderson, B.D.; Lednicky, J.A.; Torremorell, M.; Gray, G.C. The Use of Bioaerosol Sampling for Airborne Virus Surveillance in Swine Production Facilities: A Mini Review. *Front. Vet. Sci.* **2017**, *4*, 121. [\[CrossRef\]](#)

36. Anderson, B.D.; Yondon, M.; Bailey, E.S.; Duman, E.K.; Simmons, R.A.; Greer, A.G.; Gray, G.C. Environmental bioaerosol surveillance as an early warning system for pathogen detection in North Carolina swine farms: A pilot study. *Transbound. Emerg. Dis.* **2021**, *68*, 361–367. [\[CrossRef\]](#)

37. Prost, K.; Kloeze, H.; Mukhi, S.; Bozek, K.; Poljak, Z.; Mubareka, S. Bioaerosol and surface sampling for the surveillance of influenza A virus in swine. *Transbound. Emerg. Dis.* **2019**, *66*, 1210–1217. [\[CrossRef\]](#) [\[PubMed\]](#)

38. Dee, S.; Otake, S.; Oliveira, S.; Deen, J. Evidence of long distance airborne transport of porcine reproductive and respiratory syndrome virus and *Mycoplasma hyopneumoniae*. *Vet. Res.* **2009**, *40*, 39. [\[CrossRef\]](#) [\[PubMed\]](#)

39. Torremorell, M.; Pijoan, C.; Janni, K.; Walker, R.; Joo, H.S. Airborne transmission of *Actinobacillus pleuropneumoniae* and porcine reproductive and respiratory syndrome virus in nursery pigs. *Am. J. Vet. Res.* **1997**, *58*, 828–832. [\[CrossRef\]](#)

40. Nathues, H.; Spergser, J.; Rosengarten, R.; Kreienbrock, L.; Grosse Beilage, E. Value of the clinical examination in diagnosing enzootic pneumonia in fattening pigs. *Vet. J.* **2012**, *193*, 443–447. [\[CrossRef\]](#)

41. Prickett, J.; Simer, R.; Christopher-Hennings, J.; Yoon, K.J.; Evans, R.B.; Zimmerman, J.J. Detection of Porcine reproductive and respiratory syndrome virus infection in porcine oral fluid samples: A longitudinal study under experimental conditions. *J. Vet. Diagn. Investig.* **2008**, *20*, 156–163. [\[CrossRef\]](#) [\[PubMed\]](#)

42. Landis, J.R.; Koch, G.G. The Measurement of Observer Agreement for Categorical Data. *Biometrics* **1977**, *33*, 159–174. [\[CrossRef\]](#)

43. Duerlinger, S.; Knecht, C.; Sawyer, S.; Balka, G.; Zaruba, M.; Ruemenapf, T.; Kraft, C.; Rathkjen, P.H.; Ladinig, A. Efficacy of a Modified Live Porcine Reproductive and Respiratory Syndrome Virus 1 (PRRSV-1) Vaccine against Experimental Infection with PRRSV AUT15-33 in Weaned Piglets. *Vaccines* **2022**, *10*, 934. [\[CrossRef\]](#) [\[PubMed\]](#)

44. Ingram, D.L.; Dauncey, M.J. Circadian rhythms in the pig. *Comp. Biochem. Physiol. Part A Physiol.* **1985**, *82*, 1–5. [\[CrossRef\]](#)

45. Polson, D.; Berckmanns, D.; Cui, Z.; Quinn, B. Determining the optimal placement and configuration of an audio-based sensor platform to enable improved detection and characterisation of clinical respiratory episodes in large growing pig airspaces and sites. In Proceedings of the European Conference on Precision Livestock Farming, Cork, Ireland, 26–29 August 2019; pp. 490–495.

46. Neira, V.; Allerson, M.; Corzo, C.; Culhane, M.; Rendahl, A.; Torremorell, M. Detection of influenza A virus in aerosols of vaccinated and non-vaccinated pigs in a warm environment. *PLoS ONE* **2018**, *13*, e0197600. [\[CrossRef\]](#) [\[PubMed\]](#)

47. Polson, D.; Playter, S.; Berckmans, D.; Cui, Z.; Quinn, B. Evaluation of an audio-based sensor platform to classify patterns of clinical respiratory episodes in large growing pig populations. In Proceedings of the European Conference on Precision Livestock Farming, Cork, Ireland, 26–29 August 2019; pp. 553–558.

48. Biernacka, K.; Karbowiak, P.; Wrobel, P.; Chareza, T.; Czopowicz, M.; Balka, G.; Goodell, C.; Rauh, R.; Stadejek, T. Detection of porcine reproductive and respiratory syndrome virus (PRRSV) and influenza A virus (IAV) in oral fluid of pigs. *Res. Vet. Sci.* **2016**, *109*, 74–80. [\[CrossRef\]](#) [\[PubMed\]](#)

49. Kittawornrat, A.; Prickett, J.; Chittick, W.; Wang, C.; Engle, M.; Johnson, J.; Patnayak, D.; Schwartz, T.; Whitney, D.; Olsen, C.; et al. Porcine reproductive and respiratory syndrome virus (PRRSV) in serum and oral fluid samples from individual boars: Will oral fluid replace serum for PRRSV surveillance? *Virus Res.* **2010**, *154*, 170–176. [\[CrossRef\]](#)

50. Kittawornrat, A.; Panyasing, Y.; Goodell, C.; Wang, C.; Gauger, P.; Harmon, K.; Rauh, R.; Desfresne, L.; Levis, I.; Zimmerman, J. Porcine reproductive and respiratory syndrome virus (PRRSV) surveillance using pre-weaning oral fluid samples detects circulation of wild-type PRRSV. *Vet. Microbiol.* **2014**, *168*, 331–339. [\[CrossRef\]](#)

51. Kristensen, C.S.; Angen, O.; Andreasen, M.; Takai, H.; Nielsen, J.P.; Jorsal, S.E. Demonstration of airborne transmission of *Actinobacillus pleuropneumoniae* serotype 2 between simulated pig units located at close range. *Vet. Microbiol.* **2004**, *98*, 243–249. [\[CrossRef\]](#) [\[PubMed\]](#)

52. Tobias, T.J.; Bouma, A.; van den Broek, J.; van Nes, A.; Daemen, A.J.; Wagenaar, J.A.; Stegeman, J.A.; Klinkenberg, D. Transmission of *Actinobacillus pleuropneumoniae* among weaned piglets on endemically infected farms. *Prev. Vet. Med.* **2014**, *117*, 207–214. [\[CrossRef\]](#)

29. Deffner, P.; Maurer, R.; Cvjetkovic, V.; Sipos, W.; Krejci, R.; Ritzmann, M.; Eddicks, M. 横断面研究猪场中不同生产阶段支原体肺炎的发病率。兽医记录 **2022**, *191*, 1317 [[CrossRef](#)] [PubMed](#)

30. Giergiel, M.; Olejnik, M.; Jabłoński, A.; Posyniak, A. 猪口腔液中应激标志物。兽医研究杂志 2021, 65, 487–495.[\[CrossRef\]](#)

31. Gajda, A.; Jablonski, A.; Gbylik-Sikorska, M.; Posyniak, A. 猪肌肉注射土霉素后口腔液和血浆中土霉素浓度的相关性。兽医药物与治疗杂志 2017, 40, e39–e44. [\[CrossRef\]](#) [\[PubMed\]](#)

32. Tedeschini, E.; Pasqualini, S.; Emiliani, C.; Marini, E.; Valecchi, A.; Laoreti, C.; Ministrini, S.; Camilloni, B.; Castronari, R.; Patoia, L.; et al. 监测不同医院环境中室内生物气溶胶以检测 SARS-CoV-2。公共健康前沿 2023, 11, 1169073. [\[CrossRef\]](#)

33. Song, L.; Wang, C.; Jiang, G.; Ma, J.; Li, Y.; Chen, H.; Guo, J. 生物气溶胶是抗生素抗性基因在猪场中传播的重要途径。环境国际 2021, 154, 106559. [\[CrossRef\]](#)

34. Nguyen, N.T.T.; Luu, Y.T.H.; Hoang, T.D.; Nguyen, H.X.; Dao, T.D.; Bui, V.N.; Gray, G.C. 对北越工业猪场中猪链球菌的流行病学研究。One Health 2021, 13, 100254. [\[CrossRef\]](#) [\[PubMed\]](#)

35. Anderson, B.D.; Lednicky, J.A.; Torremorell, M.; Gray, G.C. 生物气溶胶采样在猪生产设施中空气传播病毒监测中的应用：一篇简短综述。兽医科学前沿 2017, 4, 121. [\[CrossRef\]](#)

36. Anderson, B.D.; Yondon, M.; Bailey, E.S.; Duman, E.K.; Simmons, R.A.; Greer, A.G.; Gray, G.C. 环境生物气溶胶监测作为北卡罗来纳州猪场病原体检测的早期预警系统：一项试点研究。跨境新兴疾病 2021, 68, 361–367. [\[CrossRef\]](#)

37. Prost, K.; Kloeze, H.; Mukhi, S.; Bozek, K.; Poljak, Z.; Mubareka, S. 生物气溶胶和表面采样用于监测猪流感 A 病毒。跨境新兴疾病 2019, 66, 1210–1217. [\[CrossRef\]](#) [\[PubMed\]](#)

38. Dee, S.; Otake, S.; Oliveira, S.; Deen, J. 猪繁殖与呼吸综合征病毒和支原体肺炎的长距离空气传播的证据。兽医研究 2009, 40, 39. [\[CrossRef\]](#) [\[PubMed\]](#)

39. Torremorell, M.; Pijoan, C.; Janni, K.; Walker, R.; Joo, H.S. 保育舍中放线杆菌胸膜肺炎和猪繁殖与呼吸综合征病毒的空气传播。美国兽医研究杂志 1997, 58, 828–832. [\[CrossRef\]](#)

40. Nathues, H.; Spergser, J.; Rosengarten, R.; Kreienbrock, L.; Grosse Beilage, E. 临床检查在诊断育肥猪地方流行性肺炎中的价值。兽医杂志 2012, 193, 443–447. [\[CrossRef\]](#)

41. Prickett, J.; Simer, R.; Christopher-Hennings, J.; Yoon, K.J.; Evans, R.B.; Zimmerman, J.J. 实验条件下猪口腔液样本中猪繁殖与呼吸综合征病毒感染的检测：一项纵向研究。兽医诊断研究杂志 2008, 20, 156–163. [\[CrossRef\]](#) [PubMed](#)

42. Landis, J.R.; Koch, G.G. 分类数据的观察者一致性测量。生物统计学 1977, 33, 159–174. [\[CrossRef\]](#)

43. Duerlinger, S.; Knecht, C.; Sawyer, S.; Balka, G.; Zaruba, M.; Ruemenapf, T.; Kraft, C.; Rathkjen, P.H.; Ladinig, A. 改良活猪繁殖与呼吸综合征病毒 1 (PRRSV-1) 疫苗对断奶仔猪中 PRRSV AUT 15 33 **2022**, *10*, 934 [[CrossRef\]](#) [PubMed](#)]

44. Ingram, D.L.; Dauncey, M.J. 猪的昼夜节律。比较生物化学与生理学 A：生理学 1985, 82, 1–5. [\[CrossRef\]](#)

45. Polson, D.; Berckmanns, D.; Cui, Z.; Quinn, B. 确定基于音频的传感器平台的最佳放置和配置，以实现改进的大型生长猪空气空间和地点中临床呼吸事件的检测和表征。欧洲精准畜牧业会议论文集，爱尔兰科克，2019 年 8 月 26–29 日；第 490–495 页。46. Neira, V.; Allerson, M.; Corzo, C.; Culhane, M.; Rendahl, A.; Torremorell, M. 在温暖环境中对接种和未接种猪气溶胶中流感 A 病毒的检测。PLOS ONE 2018, 13, e0197600. [\[CrossRef\]](#) [\[PubMed\]](#)

47. Polson, D.; Playter, S.; Berckmans, D.; Cui, Z.; Quinn, B. 评估基于音频的传感器平台对大型生长猪群体中临床呼吸事件模式的分类。欧洲精准畜牧业会议论文集，爱尔兰科克，2019 年 8 月 26–29 日；第 553–558 页。48. Biernacka, K.; Karbowiak, P.; Wrobel, P.; Chareza, T.; Czopowicz, M.; Balka, G.; Goodell, C.; Rauh, R.; Stadejek, T. 在猪口腔液中检测猪繁殖与呼吸综合征病毒 (PRRSV) 和流感 A 病毒 (IAV)。兽医研究杂志 2016, 109, 74–80. [\[CrossRef\]](#) [\[PubMed\]](#)

49. Kittawornrat, A.; Prickett, J.; Chittick, W.; Wang, C.; Engle, M.; Johnson, J.; Patnayak, D.; Schwartz, T.; Whitney, D.; Olsen, C.; et al. 来自个体公猪血清和口腔液样本中的猪繁殖与呼吸综合征病毒 (PRRSV)：口腔液将取代血清用于 PRRSV 监测吗？病毒研究 2010, 154, 170–176. [\[CrossRef\]](#)

50. Kittawornrat, A.; Panyasing, Y.; Goodell, C.; Wang, C.; Gauger, P.; Harmon, K.; Rauh, R.; Desfresne, L.; Levis, I.; Zimmerman, J. 使用断奶前口腔液样本进行猪繁殖与呼吸综合征病毒 (PRRSV) 监测可检测野生型 PRRSV 的循环。兽医微生物学 2014, 168, 331–339. [\[CrossRef\]](#)

51. Kristensen, C.S.; Angen, O.; Andreasen, M.; Takai, H.; Nielsen, J.P.; Jorsal, S.E. 在近距离放置的模拟猪单元之间展示放线杆菌胸膜肺炎血清 2 的空气传播。兽医微生物学 2004, 98, 243–249.[\[CrossRef\]](#) [\[PubMed\]](#)

52. Tobias, T.J.; Bouma, A.; van den Broek, J.; van Nes, A.; Daemen, A.J.; Wagenaar, J.A.; Stegeman, J.A.; Klinkenberg, D. 在地方流行性感染的猪场中育肥仔猪之间放线杆菌胸膜肺炎的传播。预防兽医医学 2014, 117, 207–214.[\[CrossRef\]](#)

53. Martinez-Boixaderas, N.; Garza-Moreno, L.; Sibila, M.; Segales, J. Impact of maternally derived immunity on immune responses elicited by piglet early vaccination against the most common pathogens involved in porcine respiratory disease complex. *Porcine Health Manag.* **2022**, *8*, 11. [\[CrossRef\]](#) [\[PubMed\]](#)

54. Nielsen, G.B.; Nielsen, J.P.; Haugegaard, J.; Leth, S.C.; Larsen, L.E.; Kristensen, C.S.; Pedersen, K.S.; Stege, H.; Hjulsager, C.K.; Houe, H. Comparison of serum pools and oral fluid samples for detection of porcine circovirus type 2 by quantitative real-time PCR in finisher pigs. *Porcine Health Manag.* **2018**, *4*, 2. [\[CrossRef\]](#) [\[PubMed\]](#)

55. Fachinger, V.; Bischoff, R.; Jedidia, S.B.; Saalmueller, A.; Elbers, K. The effect of vaccination against porcine circovirus type 2 in pigs suffering from porcine respiratory disease complex. *Vaccine* **2008**, *26*, 1488–1499. [\[CrossRef\]](#)

56. Kixmoeller, M.; Ritzmann, M.; Eddicks, M.; Saalmueller, A.; Elbers, K.; Fachinger, V. Reduction of PMWS-associated clinical signs and co-infections by vaccination against PCV2. *Vaccine* **2008**, *26*, 3443–3451. [\[CrossRef\]](#)

57. Pieters, M.; Daniels, J.; Rovira, A. Comparison of sample types and diagnostic methods for in vivo detection of *Mycoplasma hyopneumoniae* during early stages of infection. *Vet. Microbiol.* **2017**, *203*, 103–109. [\[CrossRef\]](#)

58. Roos, L.R.; Fano, E.; Homwong, N.; Payne, B.; Pieters, M. A model to investigate the optimal seeder-to-naïve ratio for successful natural *Mycoplasma hyopneumoniae* gilt exposure prior to entering the breeding herd. *Vet. Microbiol.* **2016**, *184*, 51–58. [\[CrossRef\]](#) [\[PubMed\]](#)

59. Otake, S.; Dee, S.; Corzo, C.; Oliveira, S.; Deen, J. Long-distance airborne transport of infectious PRRSV and *Mycoplasma hyopneumoniae* from a swine population infected with multiple viral variants. *Vet. Microbiol.* **2010**, *145*, 198–208. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.

53. Martinez-Boixaderas, N.; Garza-Moreno, L.; Sibila, M.; Segales, J. Impact of maternally derived immunity on immune responses elicited by piglet early vaccination against the most common pathogens involved in porcine respiratory disease complex. *Porcine Health Manag.* **2022**, *8*, 11. [\[CrossRef\]](#) [\[PubMed\]](#)

54. Nielsen, G.B.; Nielsen, J.P.; Haugegaard, J.; Leth, S.C.; Larsen, L.E.; Kristensen, C.S.; Pedersen, K.S.; Stege, H.; Hjulsager, C.K.; Houe, H. Comparison of serum pools and oral fluid samples for detection of porcine circovirus type 2 by quantitative real-time PCR in finisher pigs. *Porcine Health Manag.* **2018**, *4*, 2. [\[CrossRef\]](#) [\[PubMed\]](#)

55. Fachinger, V.; Bischoff, R.; Jedidia, S.B.; Saalmueller, A.; Elbers, K. The effect of vaccination against porcine circovirus type 2 in pigs suffering from porcine respiratory disease complex. *Vaccine* **2008**, *26*, 1488–1499. [\[CrossRef\]](#)

56. Kixmoeller, M.; Ritzmann, M.; Eddicks, M.; Saalmueller, A.; Elbers, K.; Fachinger, V. Reduction of PMWS-associated clinical signs and co-infections by vaccination against PCV2. *Vaccine* **2008**, *26*, 3443–3451. [\[CrossRef\]](#)

57. Pieters, M.; Daniels, J.; Rovira, A. Comparison of sample types and diagnostic methods for in vivo detection of *Mycoplasma hyopneumoniae* during early stages of infection. *Vet. Microbiol.* **2017**, *203*, 103–109. [\[CrossRef\]](#)

58. Roos, L.R.; Fano, E.; Homwong, N.; Payne, B.; Pieters, M. A model to investigate the optimal seeder-to-naïve ratio for successful natural *Mycoplasma hyopneumoniae* gilt exposure prior to entering the breeding herd. *Vet. Microbiol.* **2016**, *184*, 51–58. [\[CrossRef\]](#) [\[PubMed\]](#)

59. Otake, S.; Dee, S.; Corzo, C.; Oliveira, S.; Deen, J. Long-distance airborne transport of infectious PRRSV and *Mycoplasma hyopneumoniae* from a swine population infected with multiple viral variants. *Vet. Microbiol.* **2010**, *145*, 198–208. [\[CrossRef\]](#)

免责声明/出版商注：所有出版物中的声明、观点和数据均仅为个人作者和贡献者的个人观点，而非MDPI和/或编辑的观点。MDPI和/或编辑不承担因内容中提到的任何思想、方法、说明或产品而导致的任何人员或财产损失的责任。